Discrete Response, Time Series and Panel Data - Notes

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W-01-Timeseries-Introduction-Visualisation-Stationarity

> **Note**
>
> Coming soon...

W-02-Timeseries-Introduction-Visualisation-Stationarity

Log-transformations

Instead of the original time series , consider the log-transformed time series with the elements.

$$g_t = \log(X_t)$$

When to apply it?:

- When the distribution of the time series elements is right-skewed.
- When a time series could be produced by the product of model outcomes.

Differencing

Let's take a linear function (defined on a set of integer numbers)

$$X_t = m \cdot t + b$$

Then, the backward difference

$$X_t - X_{t-1} = (m \cdot t + b) - (m \cdot (t-1) + b) = m$$

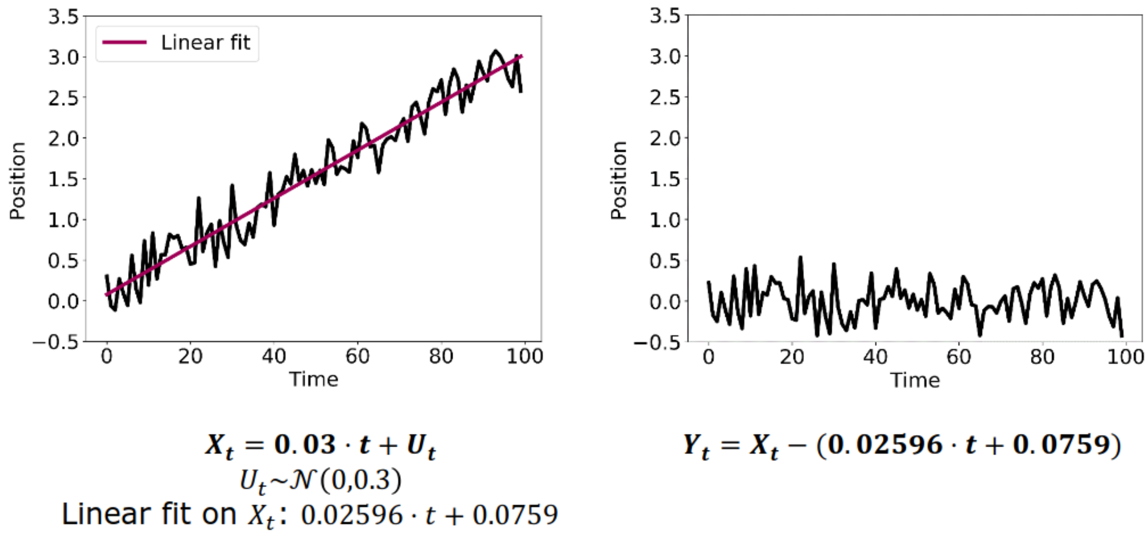


Figure 1: Differencing

Higher Order Differencing

Use multiple derivatives to eliminate lower-order terms. Trend order is reduced, but the stationary part may be more complex. After trend is eliminated, maybe a new dependencies in the stationary part.

Example

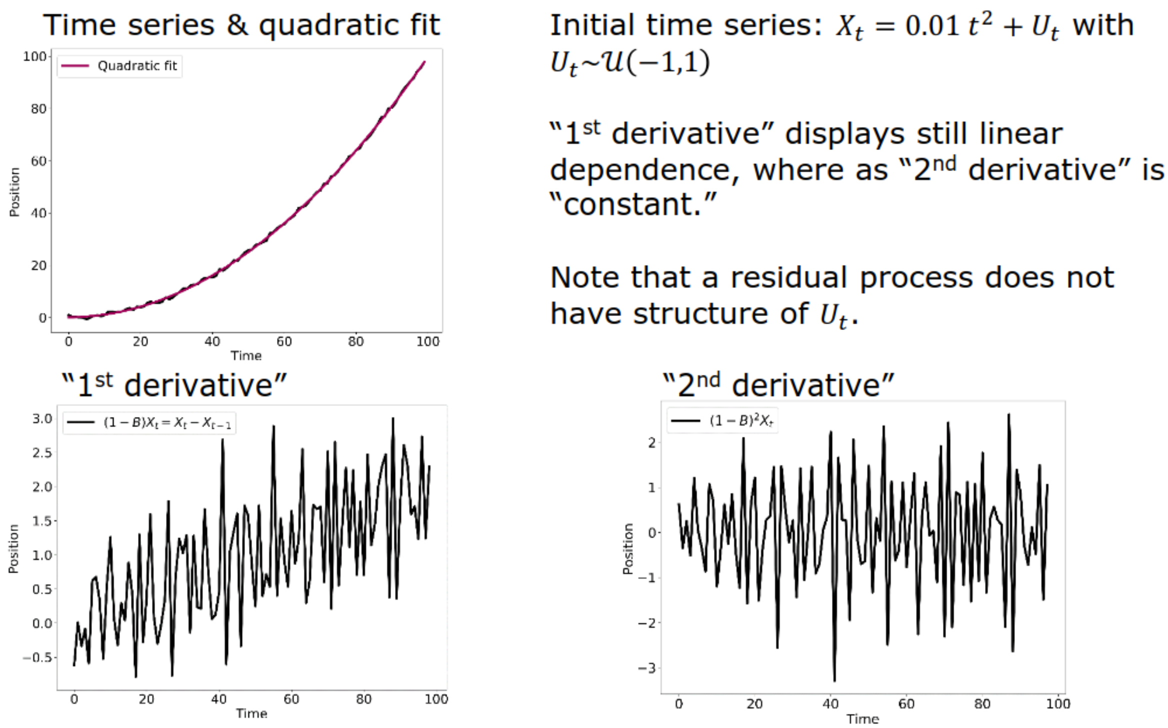
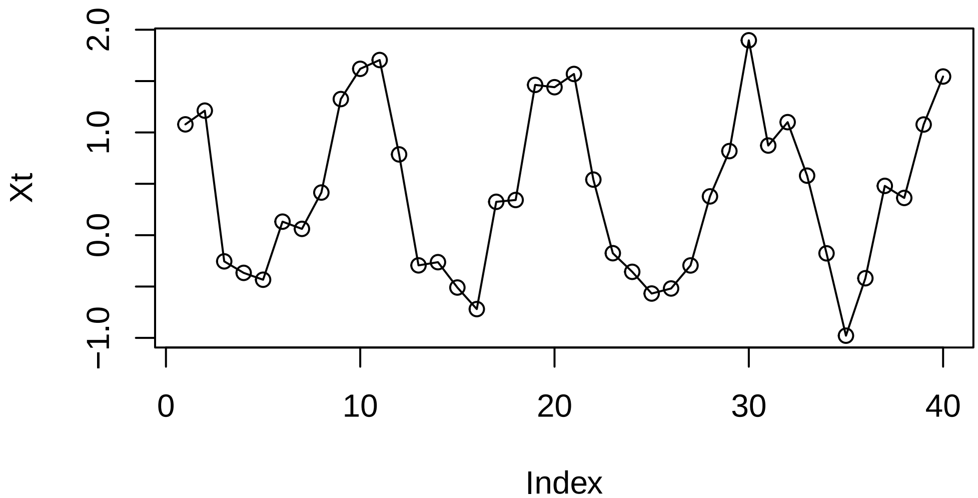


Figure 2: Remove Higher Order Trend

Diferencing Seasonality

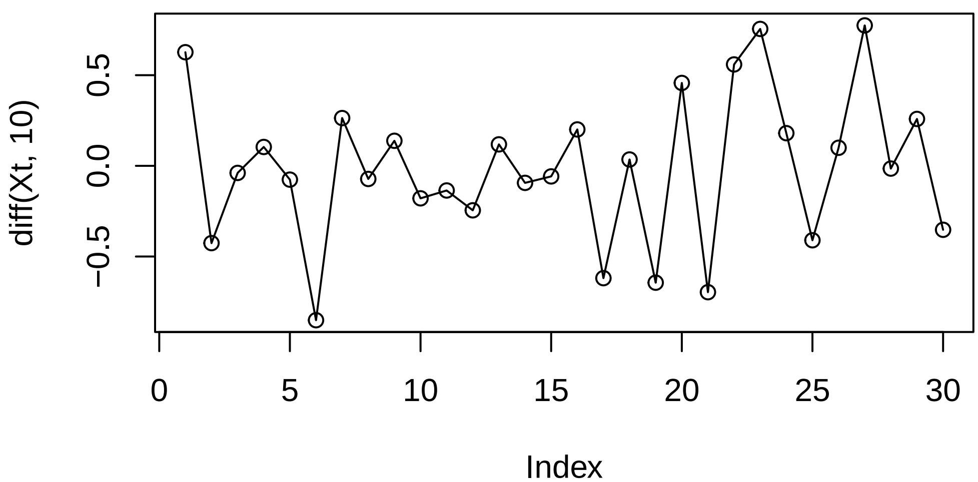
Seasonality manifests itself by a repetitive behavior after a fixed lag time p.

```
tRange<-1:40
Xt<-cos(2*pi/10*tRange)+runif(40)
plot(Xt,type="o")
```



What happens if we differentiate now with lag 10?

```
plot(diff(Xt,10),type="o")
```



Summary of differencing

Advantages:

- can remove trends and seasonality effects
- is easy to use

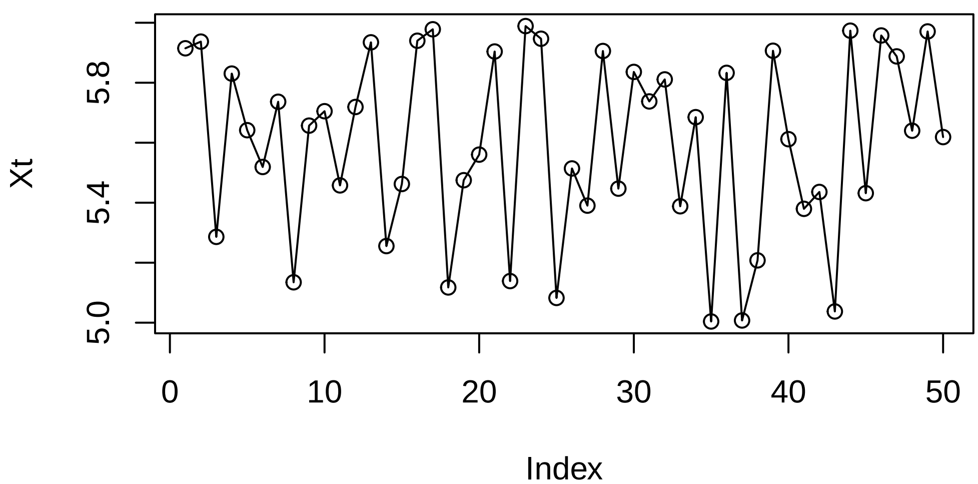
Disadvantages:

- the resulting time series is shorter than the original one
- the results of differencing may display dependencies among different elements
- the original decomposition elements are unknown

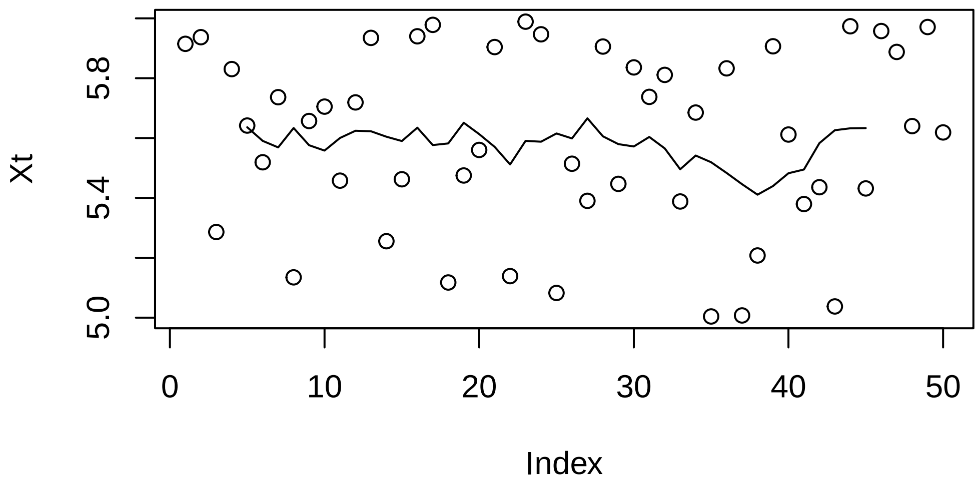
Filtration

To reduce the noise element in an experiment, you can repeat the measurement multiple times and then calculate the average.

```
set.seed(42)
Xt<-5+runif(50)
plot(Xt,type="o")
```



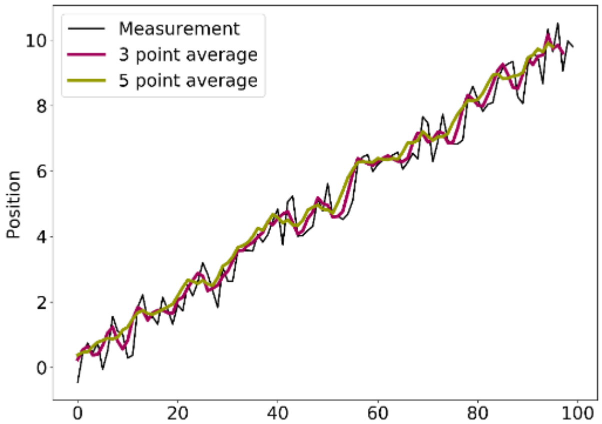
```
filtered<-filter(Xt,filter=rep(1,10)/10)
plot(Xt)
points(filtered,type="l")
```



Reasons for using Filtration

By averaging measurements, we obtain a new averaged time series. By increasing the number of averaged measurements, the parts converges to the expectation value of the measurement error, i.e. 0.

Note: The short-term variations are smoothed out and the trend is confirmed.



Initial time series:

$$X_t = 0.1\,t + U_t, U_t \sim \mathcal{U}(-1,1)$$

3 point average: $a_0 = a_{-1} = a_{-2} = \frac{1}{3}$

5 point average: $a_0 = a_{-1} = a_{-2} = a_{-3} = a_{-4} = \frac{1}{5}$

Figure 3: Filtering

Linear filters to quantify seasonality effects

In case of seasonality effects, the filter has to expand over the full period to extract the trend. To quantify the seasonal impact, the values of the same time in the season have to be averaged.

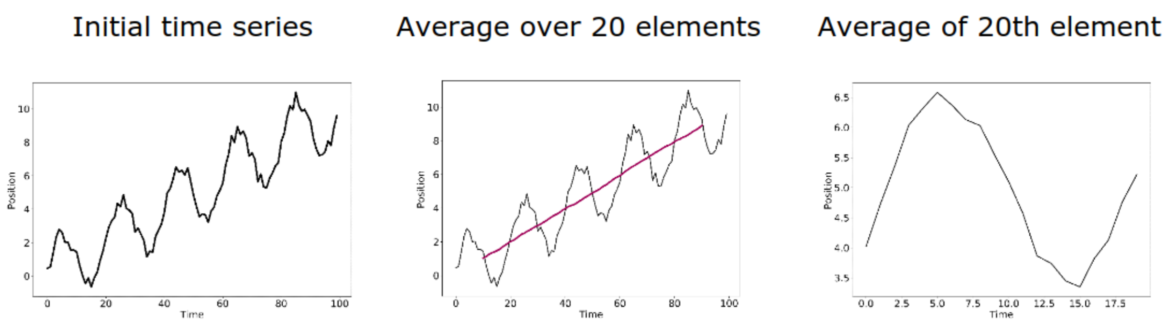


Figure 4: Quantify Seasonality Effects

Seasonal-trend decomposition by Loess Algorithm (STL)

Linear filters are great if periodicity is obvious; but in less obvious cases, extracting the individual components may be laborious. An alternative is the seasonal-trend decomposition procedure by Loess: An iterative, non-parametric smoothing algorithm that is more robust.

Fitting of Periodic Function

If you know the generative model, you can use fitting to determine the contributions of trend and seasonality. Fitting methods (among others):

- Ordinary least square (OLS)
- Robust fitting (to prevent the impact of outliers)
- GLS (generalised least square) / time series regression methods

```r
set.seed(42)
#Generate the sample data set
t<-seq(1,100,length=100)
data<-0.1*t+cos(2*pi/10*t)+runif(100)
ts<-ts(data)
#Fit the model
fit<-lm(ts~t+cos(2*pi/10*t))
summary(fit)
```

```
Call:
lm(formula = ts ~ t + cos(2 * pi/10 * t))

Residuals:
     Min       1Q   Median       3Q      Max 
-0.58266 -0.22271  0.04478  0.25411  0.49285 

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)    
(Intercept)        0.639744   0.059840   10.69   <2e-16 ***
t                  0.097718   0.001029   94.98   <2e-16 ***
cos(2 * pi/10 * t) 0.973247   0.041999   23.17   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2969 on 97 degrees of freedom
Multiple R-squared:  0.9901,    Adjusted R-squared:  0.9899 
F-statistic:  4836 on 2 and 97 DF,  p-value: < 2.2e-16
```

Autocorrelation

The autocorrelation depends only on the lag   for weakly or strongly stationary time series.

$$p(k) := Cor(X_{t+k}, X_t)$$

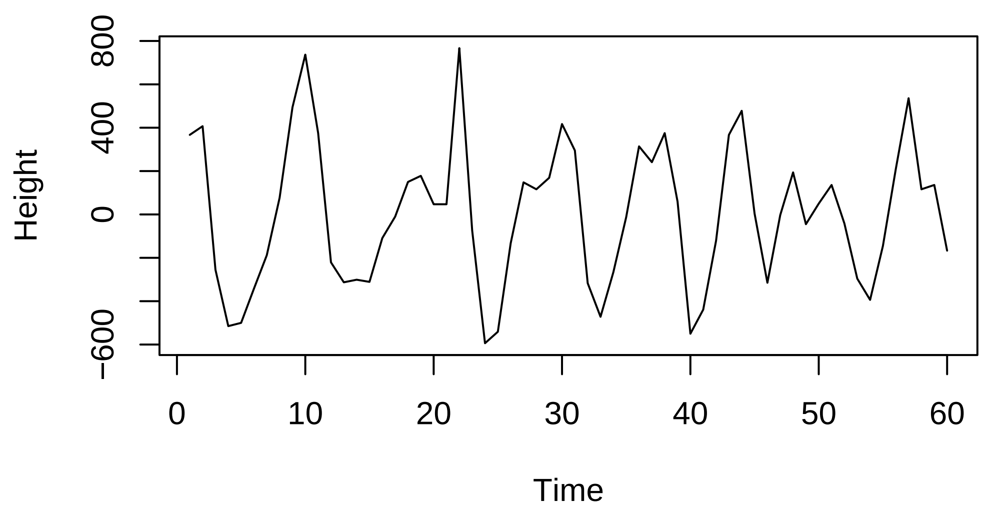
Interpretation of autocorrelation

The correlation $p(k)^2$ squared corresponds to the percentage of variability explained by the linear association between X_t and X_{t+k}.

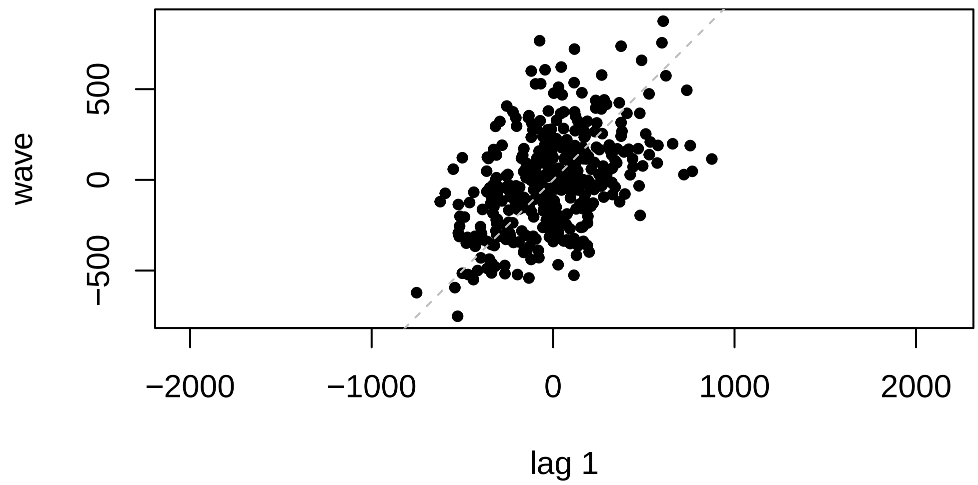
Lagged Scatter Plot

Exchange average over realizations with average over time.

```
address <- "https://raw.githubusercontent.com/AtefOuni/ts/master/Data/wave.dat"
dat <- read.table(address,header=T)

#Generate ts object
wave<-ts(dat$waveht)

#Visualize the data
plot(window(wave,1,60),ylab="Height")
```

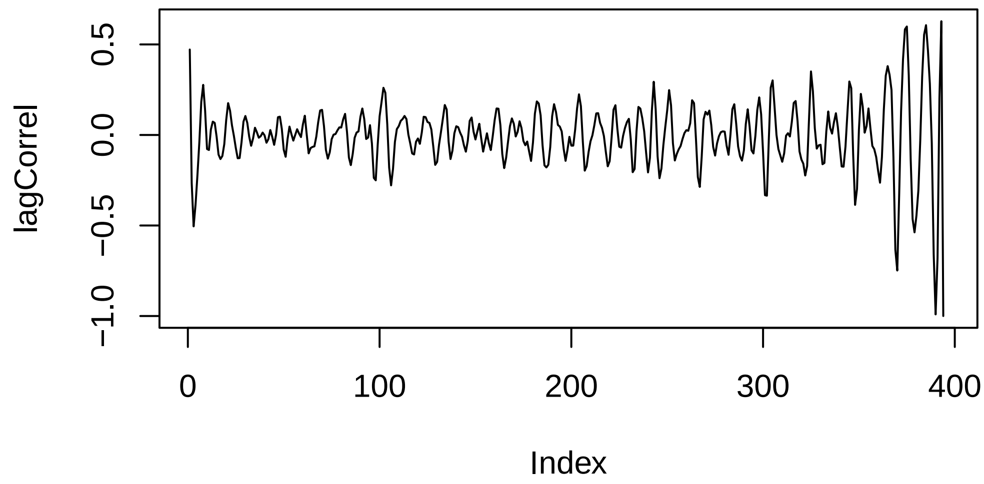


```r
#Lag plot
lag.plot(wave,do.lines=FALSE,pch=20)
```



Calculate lagged scatter plot estimator

```r
# Calculate the Pearson Correlation coefficients
n <- length(wave)
lagCorrel <- rep(0,n)

for (i in 1:(n-1)){
lagCorrel[i]=cor(wave[1:(n-i)],wave[(i+1):n])
}

plot(lagCorrel,type="l")
```



Plug-In Estimation

The plug-in estimator is the standard approach to computation autocorrelations. Calculated with an `acf` function in R.

Summary Autocorrelation

- (Auto)correlation measures the linear association at a given lag
- For realizations, the autocorrelation is estimated by running the calculations over time instances instead of realizations (therefore stationarity is desired)
  - Lagged scatter plot estimator
  - Plug-in estimator
- For high lags, the lagged scatter plot estimator diverges (because too few time points are considered)
- The plug-in estimator corrects the diverging behavior but generates a bias.

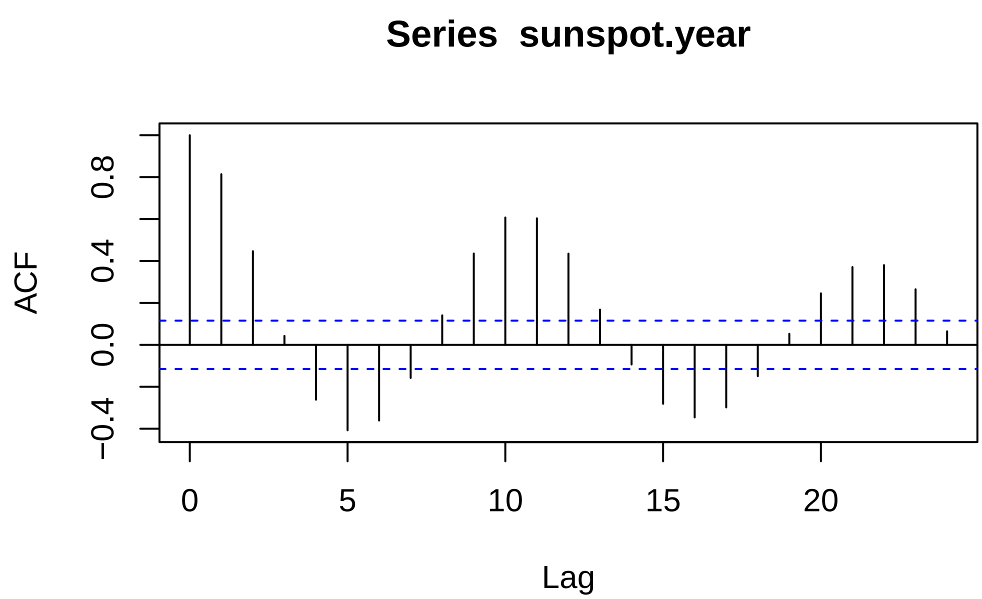
> ⚠ Warning
>
> Warning: Non-linear relations may confuse the autocorrelation.

W-03-Timeseries-ACF-AR

Correlogram

Standard command: `acf(data)` for plug-in estimator values.

```
data(sunspot.year)
acf(sunspot.year)
```



> Caution
>
> The first lag at lag 0 always have to be a value of 1

Bounds of Autocorrelation

Under the null hypothesis of a time series process with iid distributed random variables, a 95 % acceptance region for the null hypothesis is:

$$\pm\frac{1.96}{\sqrt{n}}$$

For stationary processes, plug-in estimator value $\hat{p}(k)$ within the confidence band $\pm\frac{1.96}{\sqrt{n}}$ are considered to be different from 0 only by chance.

Ljung-Box Test

The Ljung-Box test assesses the null hypothesis that the random variables are independent when the h autocorrelation coefficients $\hat{p}(k)$ for $k = 1, \dots, h$ are equal to zero.

```
Box.test(sunspot.year, lag=10, type="Ljung-Box")
```

```

	Box-Ljung test

data:  sunspot.year
X-squared = 542.41, df = 10, p-value < 2.2e-16
```

Characteristic Patterns in ACF

Patterns in ACF for non-stationary time series

For non-stationary time series, the ACF decays only slowly.

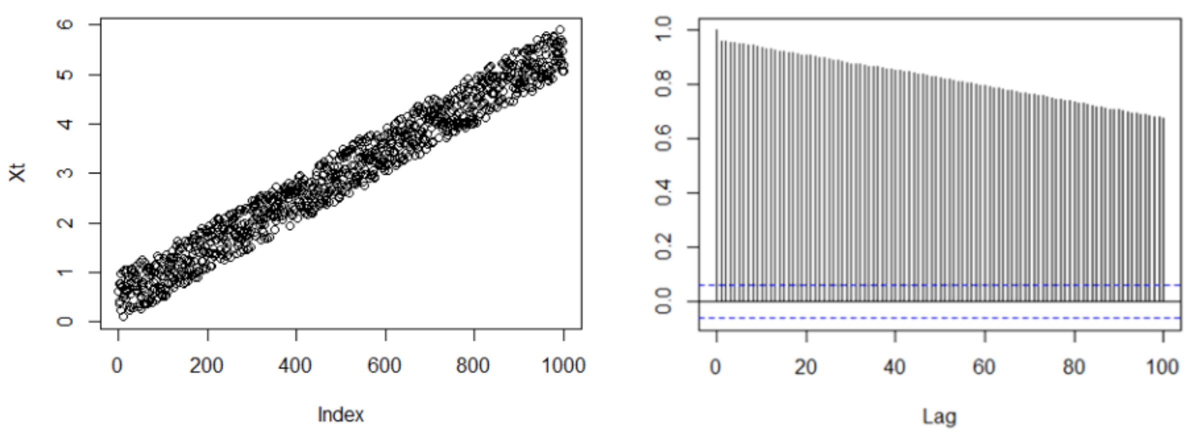


Figure 5: Non-Statiionary time seires

ACF of time series with seasonal pattern

For time series with seasonality, the ACF displays periodic structure.

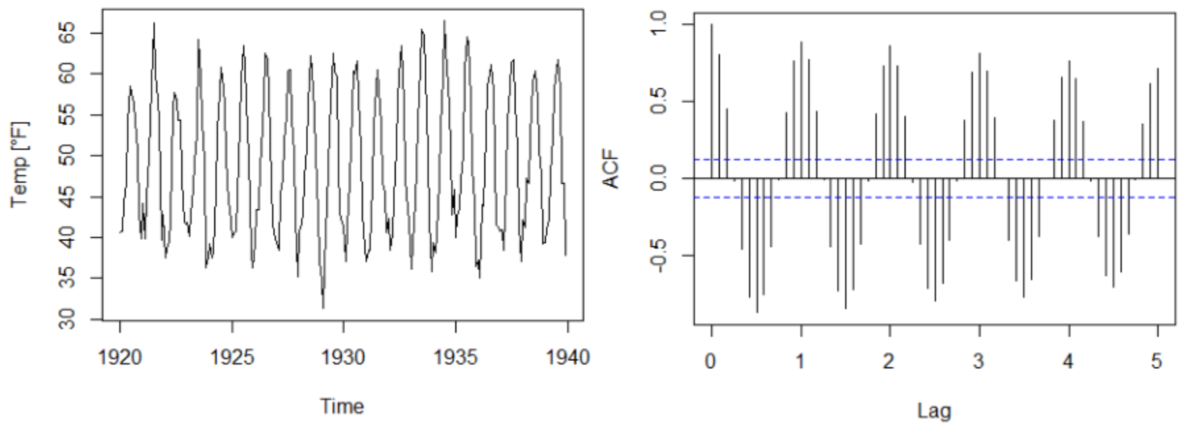


Figure 6: Time series with seasonal pattern

ACF of time series with seasonal pattern and trend

Time series with seasonality and trend, the ACF displays periodicity and slow decay.

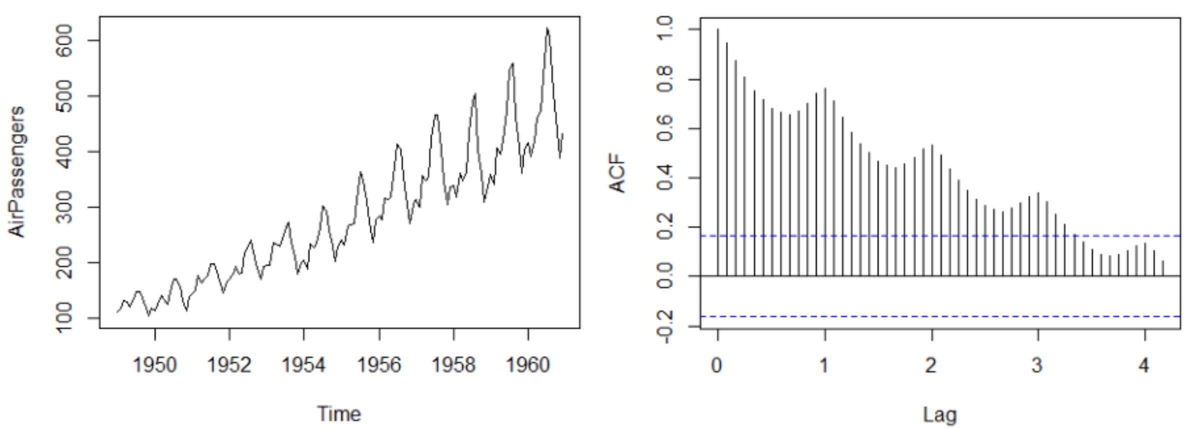


Figure 7: Time series with seasonal pattern and trend

ACF of time series with outliers

Outliers may be diagnosed by lagged scatter plots. Note that each outlier causes other outliers.

❗ Important

The plug-in estimator $\hat{p}(k)$ is sensitive to outliers.

Properties of ACF

Autocorrelation of process vs. estimated values of realization

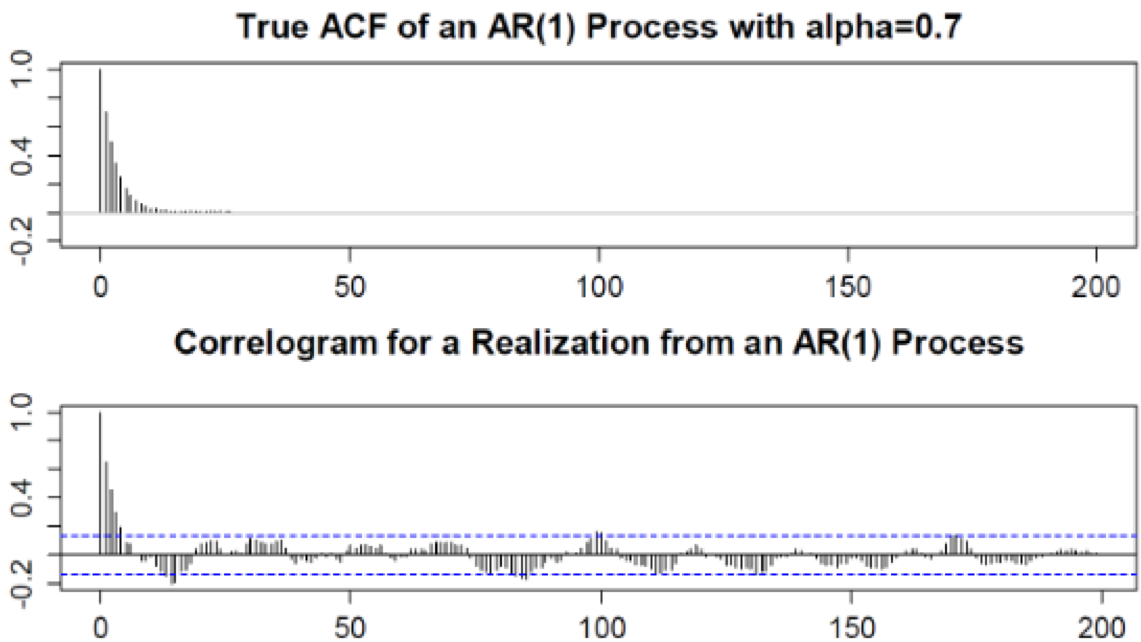
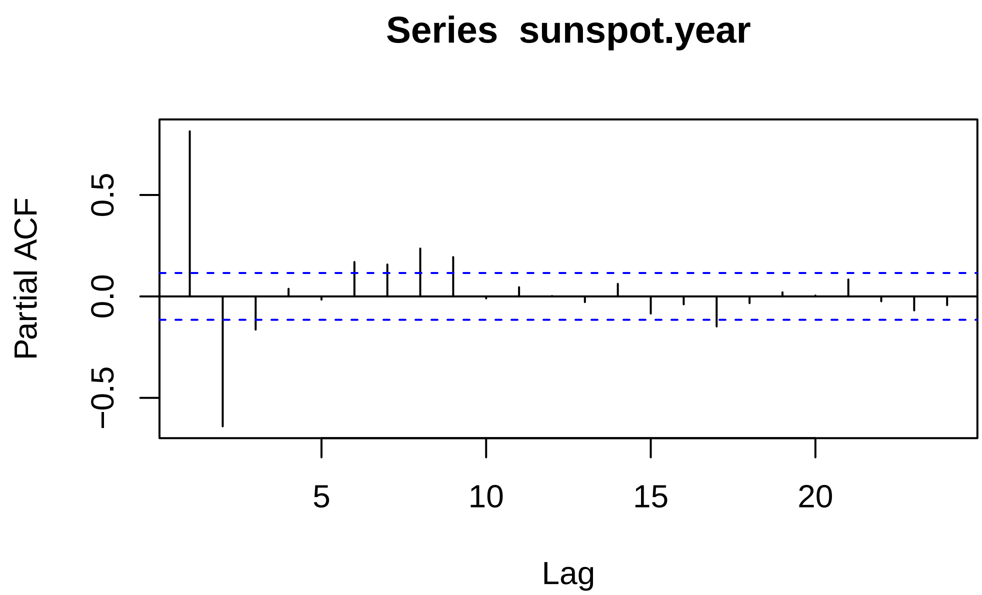


Figure 8: Correlogram of one realization compare with the true plot

Partial Autocorrelation

A challenge in the interpretation of ACFs is that the effect of smaller lags remains. The partial autocorrelation function (PACF) gives the partial correlation of a stationary time series with its own lagged values, regressed the values of the time series at all shorter lags.

```
pacf(sunspot.year)
```



W-04-Timeseries-ACF-AR

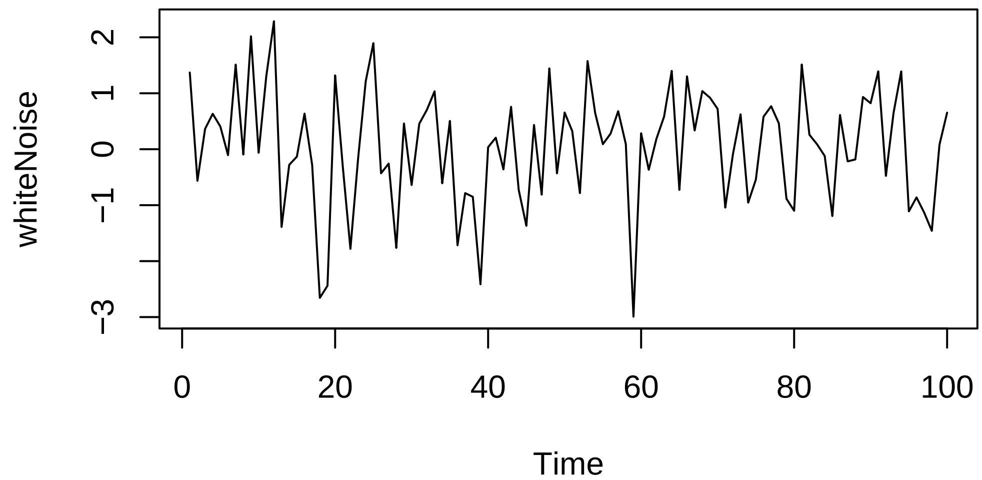
Fundamentals of Modeling

White noise

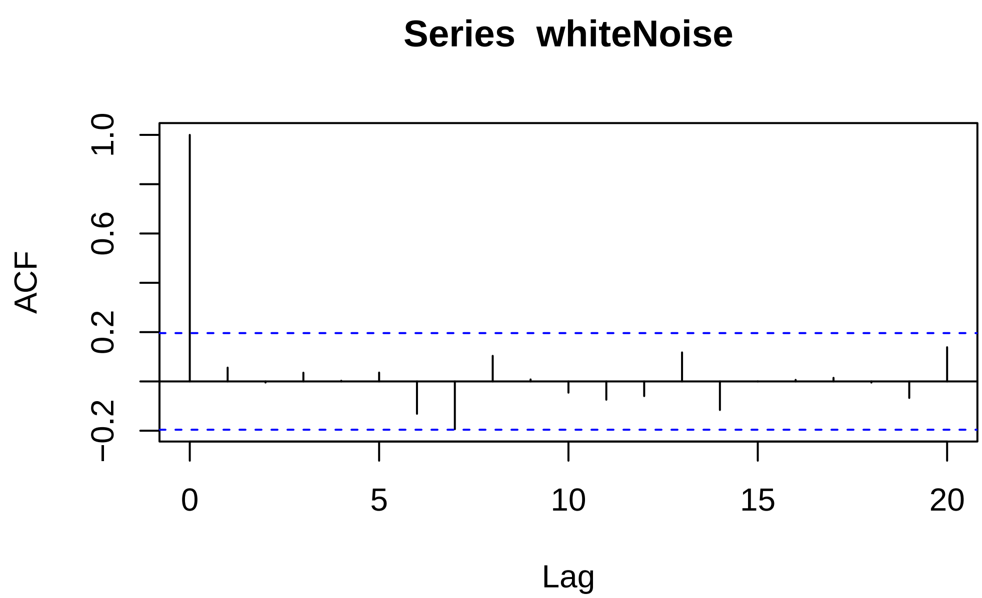
A time series process $Wt, t \in \mathbb{N}$ is called white noise if it is a sequence of independent and identically distributed random variables with mean 0.

> Note: The name "white" indicates that all frequencies are included, as in the case of "white light."

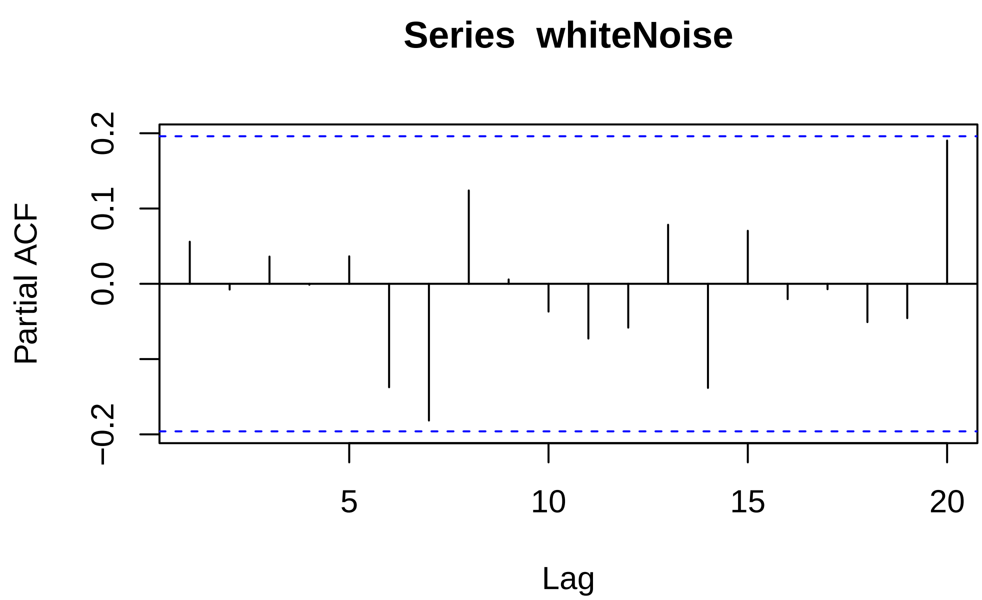
```
set.seed(42)
whiteNoise <- ts(rnorm(100,mean=0,sd=1))
plot(whiteNoise)
```




```
acf(whiteNoise)
```



```r
pacf(whiteNoise)
```



Causal White Noise

A series of *iid* random variables e_t, which are individually independent of the proceeding time series elements X_t

Basic Time Series Models

The following models are often used when there is limited information about how time series data is generated:

- AR: auto-regressive model
- MA: moving average model
- ARMA: combination of AR and MA model
- ARIMA: non-stationary ARMA model (I=integration)
- SARIMA: seasonal ARIMA model

Auto-Regression Models (AR)

Express the current random variable as a linear combination of the last p time steps and a "completely independent" term e_t called innovation.

$$X_t = \alpha_1 X_{t-1} + \cdots + \alpha_p X_{t-p} + e_t$$

where α_x is a set of finite parameters and e_t a series of independent random variables.

$AR(1)$ as Random Walk Model

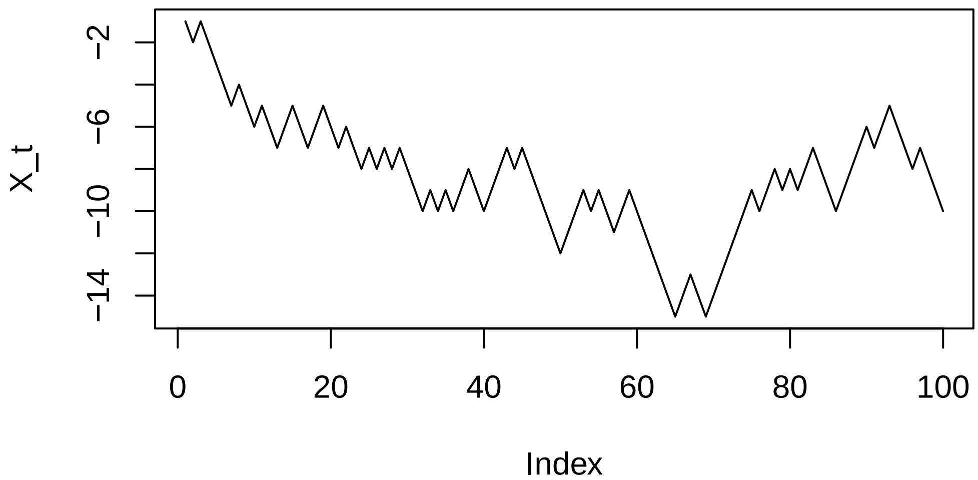
Trajectory of an object moving along a line. At each moment, the object takes randomly the decision in which direction to move on.

$$X_t = X_{t-1} + e_t$$

where e_t is random with equal probability for -1 and 1.

```
set.seed(42)
step<- 1 - 2 * round(runif(100))
X_t<-cumsum(step)
plot(X_t, type="lines")
```

```
Warning in plot.xy(xy, type, ...): plot type 'lines' will be truncated to first
character
```

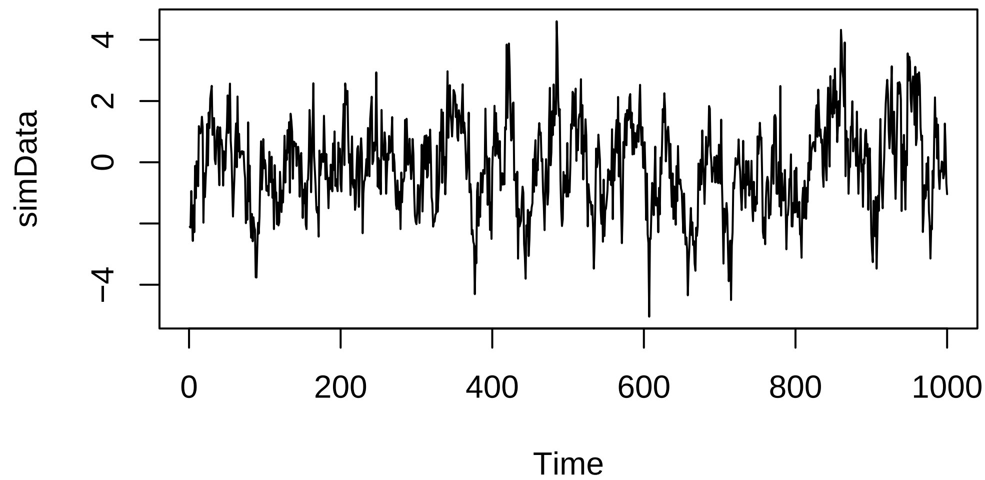


Simulation $AR(p)$ Processes

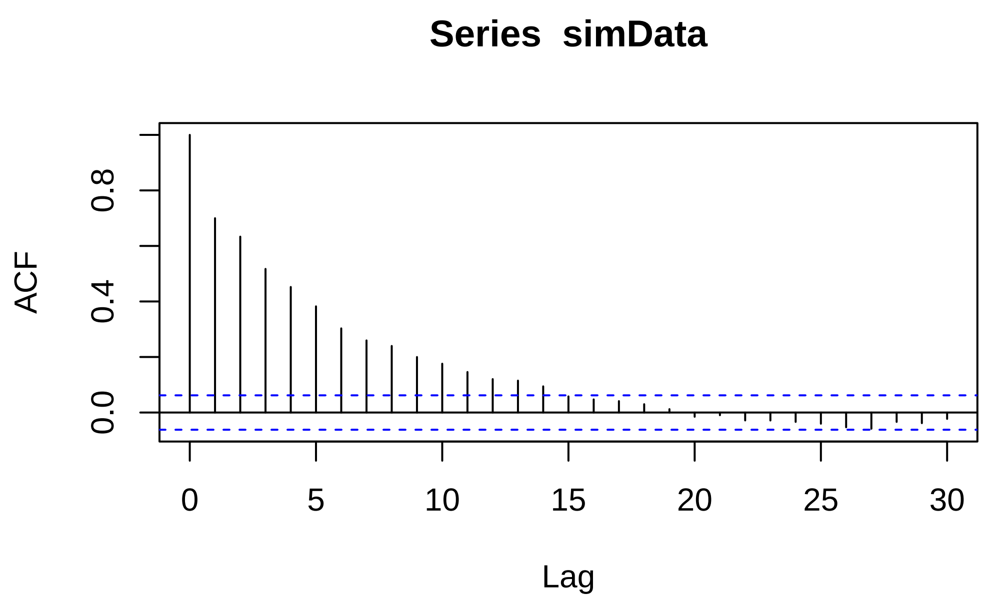
We would like to simulate:

$$X_t = 0.5X_{t-1} + 0.3X_{t-2} + e_t \text{ with } e_t \sim N(0,1) \text{ iid.}$$

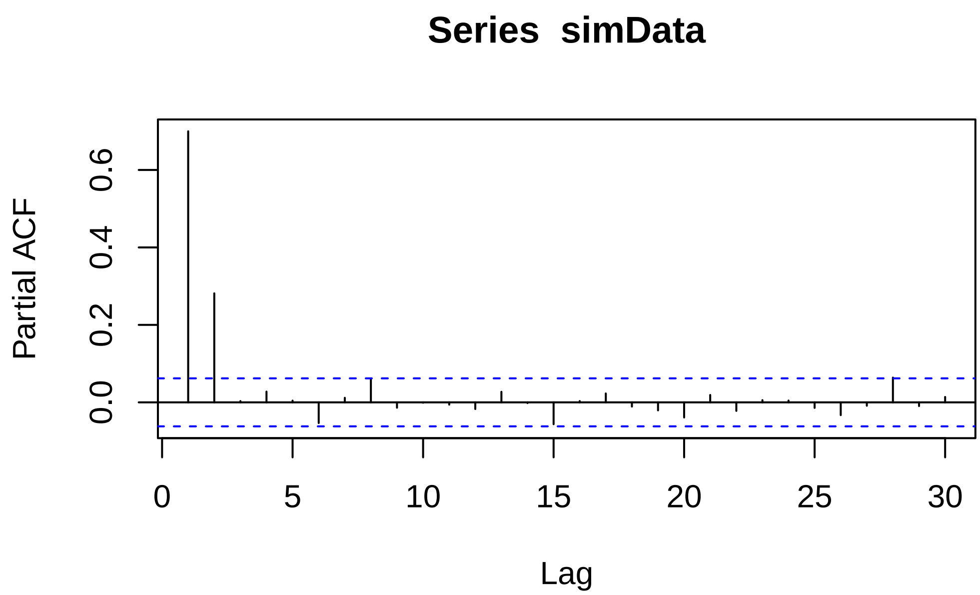
```r
set.seed(42)
ar.coefficients <- c(0.5, 0.3)
simData <- arima.sim(n=1000,model=list(ar=ar.coefficients))
plot(simData)
```




```r
acf(simData)
```



```r
pacf(simData)
```



Stationarity of $AR(p)$ Models

AR(p) models are stationary when the:

- mean is 0
- absolute value of the roots of the characteristic polynomial is all larger than 1

```r
polyroot(c(1,-0.8,-0.4))
```

```
[1]  0.8708287+0i -2.8708287+0i
```

As the first value has an absolute value below 1, the time series is not stationary.

Key Properties of $AR(p)$ Models

The AR(1) process:

$$X_t = 0.75 * X_t + e_t$$

is stationary because the characteristic polynomial:

```
polyroot(c(1,-0.75))
```

```
[1] 1.333333+0i
```

For the model identification of AR(p) processes, two properties are key:

- Autocorrelation decays quickly or displays a damped sinusoid
- Partial autocorrelation function has a clear cut-off at p

Model Order Identification

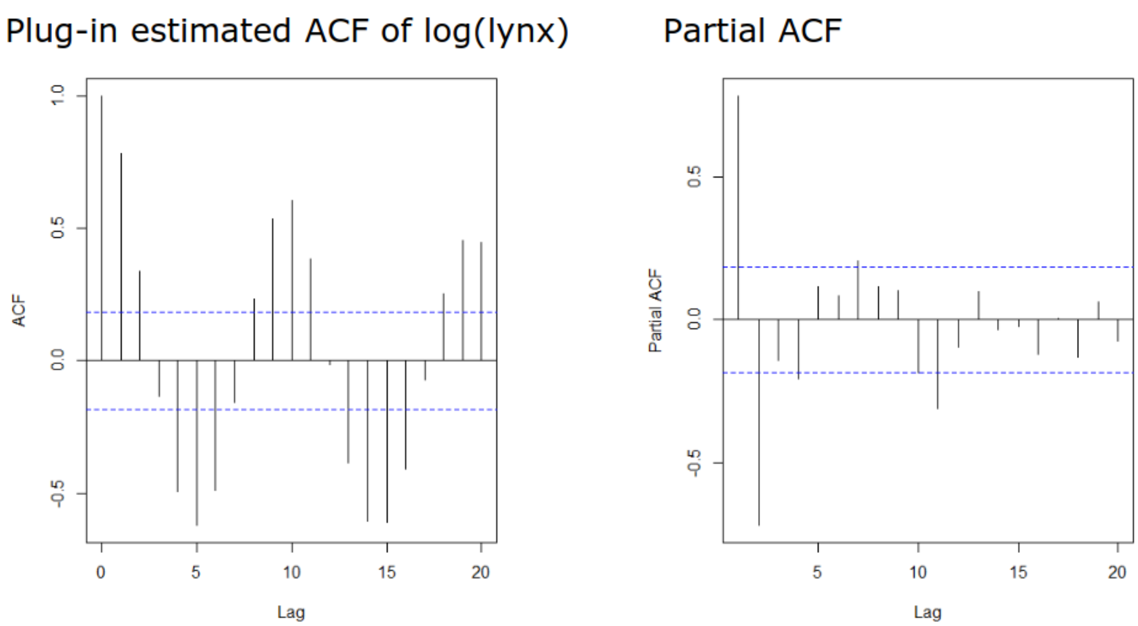


Figure 9: Plug-in estimated ACF and PACF

- ACF displays damped sinusoid
- PACF has clear cut-off after lag 11, 7, 4 or 2 (depending on the threshold).

Try fitting an $AR(2)$, $AR(4)$, $AR(7)$, $AR(11)$

W-05-Timeseries-AR-MA

Model Parameter Identification

Time series are often not centered: I.e. instead of a classical AR(p) process, a centered version has to be fitted.

$$Y_t = m + X_t = m + \alpha_1 X_{t-1} + \dots + \alpha_p X_{t-p} + e_t$$

Fitting parameters are: - the global mean m - the AR-coefficients $\alpha_1, \dots, \alpha_p$ - the parameters defining the distribution of the innovation e_t. Here, we assume a normal distribution.

Model Parameter from Ordinary Least Square (OLS)

We want to estimate the parameters from the model equation, we have a total od $n-p$ points for this, because we have a total of n time segments and each equation links p time instance.

The OLS procedure is:

1. Estimate the global mean $\hat{m} = \bar{y} = \frac{1}{n}\sum_{t=1}^{n} y_t$ and determine $x_t = y_t - \hat{m}$
2. Carry out a regression on x_t without intercept. The resulting parameters are $\hat{\alpha}, \dots, \hat{\alpha_p}$
3. Estimate the standard deviation of the innovation $\hat{\sigma}_E^2$ deviation of the residual.

```
ar.ols(data,order=p)
```

Yule-Walker Equations

Basic idea: The model coefficients and the ACF values are entangled. The values of the auto-correlation function depend on the model coefficients. A careful calculation for an AR(p) model yields the following equations, called Yule-Walker equations:

$$\sum_{i=1}^{p} \alpha_i \rho(k-i) = \rho(k) \quad \text{for } k > 0$$

$$\sum_{i=1}^{p} \alpha_i \rho(i) = \rho(0) - \sigma_E^2 \quad \text{for } k = 0$$

> Note: Note that the autocorrelation function is symmetric, i.e. $p(-k) = p(k)$ for all k.

To fit the AR(p) model parameters, we must solve the above equations with the estimated values of the auto correlation function.

```
ar.yw(data,order=p)
```

Further options for fitting

Burg's algorithm

Offers another fitting cost function that exploit also the first p function values.

```
ar.burg(data,order.max=p)
```

Maximum likelihood estimator (MLE)

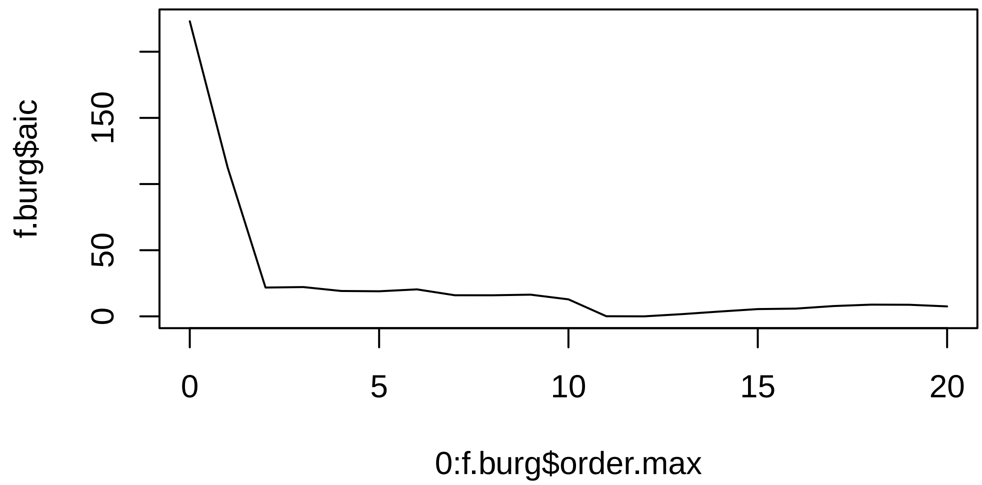
Determines the parameter based on an optimization of the maximum likelihood function

```
arima(Data,order=c(2,0,0))
```

Which method to use when?

One option is to use the Akaike information criterion (AIC), which estimates the relative amount of information lost by the considered model. The less information a model loses, the higher the quality of that model.

```
f.burg<-ar.burg(log(lynx))
plot(
    0:f.burg$order.max,
    f.burg$aic,
    type="l"
    )
```

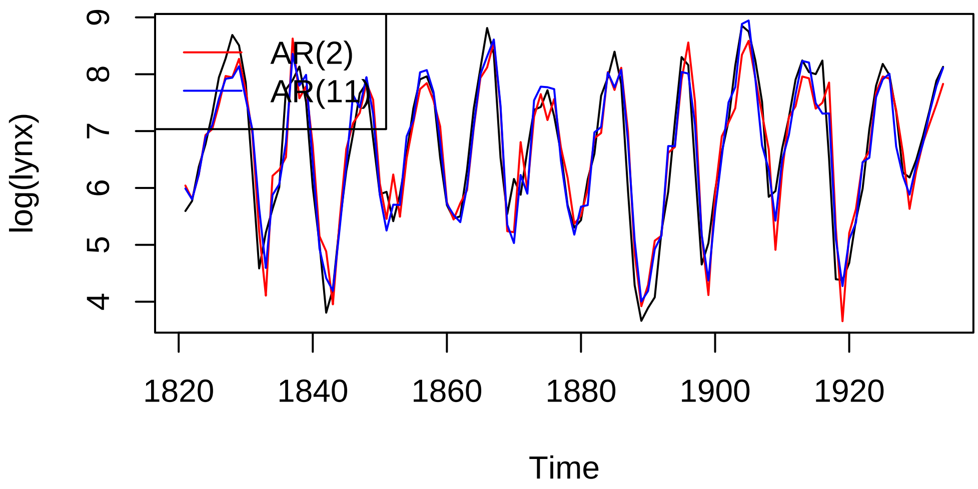


Note: No order parameter is given any more.

Model Diagnostic

Example

```
# Fitting the two models
fit.2<-arima(log(lynx),order=c(2,0,0))
fit.11<-arima(log(lynx),order=c(11,0,0))

# Display the time series and model predictions
plot(log(lynx))
lines(log(lynx) - fit.2$resid, col = "red")
lines(log(lynx) - fit.11$resid, col = "blue")
legend("topleft",
       legend = c("AR(2)", "AR(11)"),
       col = c("red", "blue"),
       lty = 1)
```

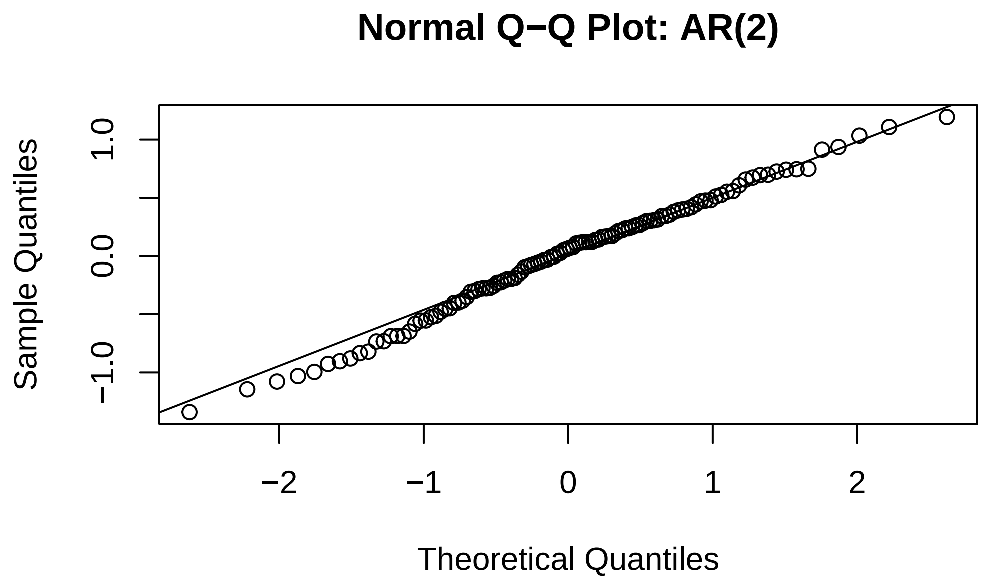


QQ-Plot

Compare quantile of measurement data with theoretical distribution

- X-Axis: Theoretical distribution
- Y-Axis: Actual data distribution

```
qqnorm(fit.2$residuals,main="Normal Q-Q Plot: AR(2)")
qqline(fit.2$residuals)
```



Moving Average Models (MA)

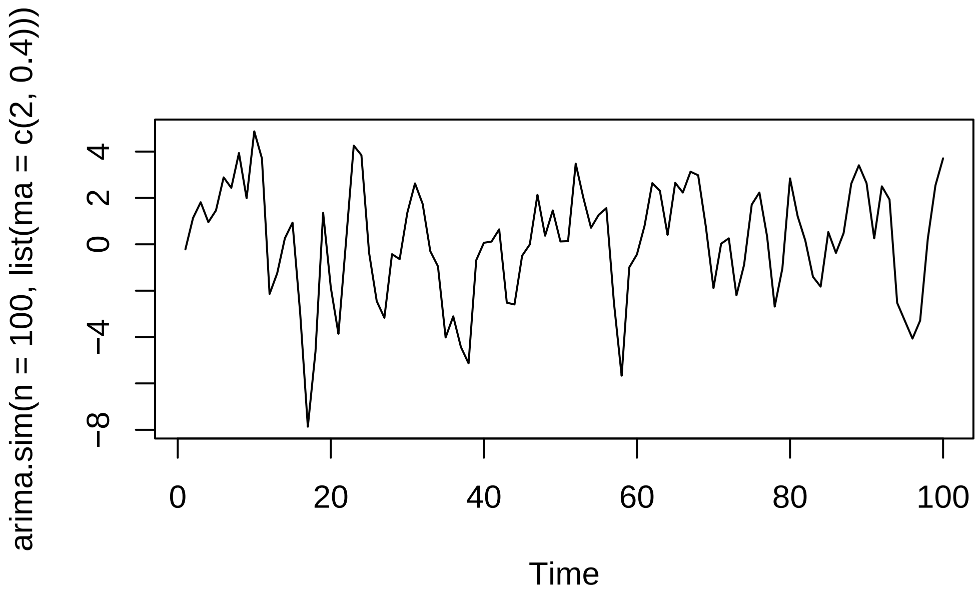
A moving average MA(q) model is a model in the form

$$X_t = E_t + \beta_1 E_{t-1} + \beta_2 E_{t-2} + \cdots + \beta_q E_{t-q}$$

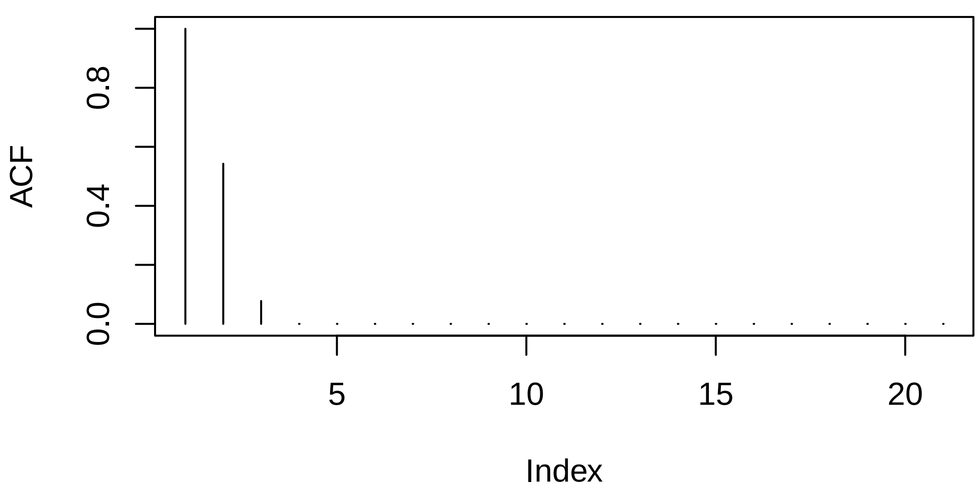
with the time series E_t of innovations.

> **Note**
>
> A moving average model does not have an explicit reference to previous values; only the innovation parts are considered.

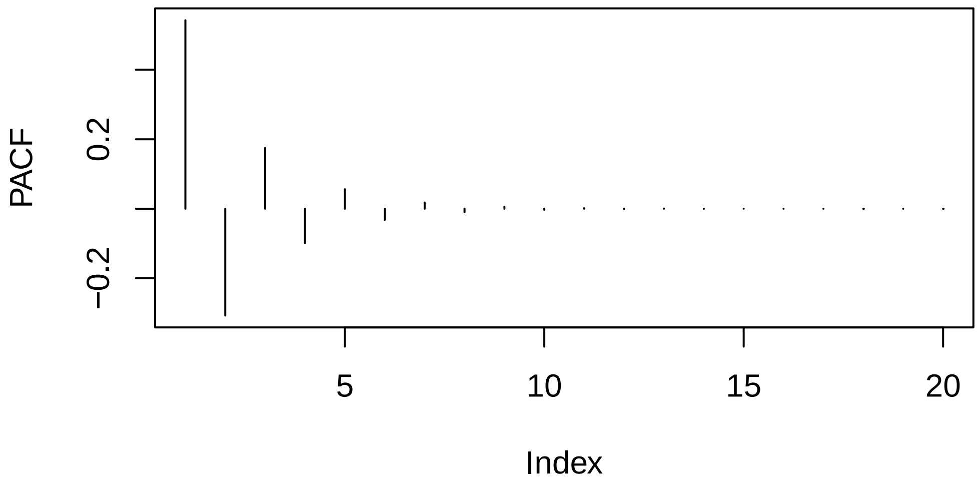
```r
set.seed(42)

#MA(2)
plot(arima.sim(n=100,list(ma=c(2,0.4))))
```




```
plot(ARMAacf(ma=c(2,0.4),lag.max=20),type="h",ylab="ACF")
```



```r
plot(ARMAacf(ma=c(2,0.4),pacf=T,lag.max=20),type="h",ylab="PACF")
```



Generalization: In general, we can show that the autocorrelation function of an $MA(q)$ vanishes for all lags larger than q.

W-06-Timeseries-Diagnostics-MA

Moving Average Model

A moving average MA(q) model is a model in the form

$$X_t = E_t + \beta_1 E_{t-1} + \beta_2 E_{t-2} + \cdots + \beta_q E_{t-q}$$

with the time series E_t of innovations. The ACF displays cutoff and PACF displays fast decay. In general, we can show that the autocorrelation function of an $MA(q)$ vanishes for all lags larger than q.

Invertibility

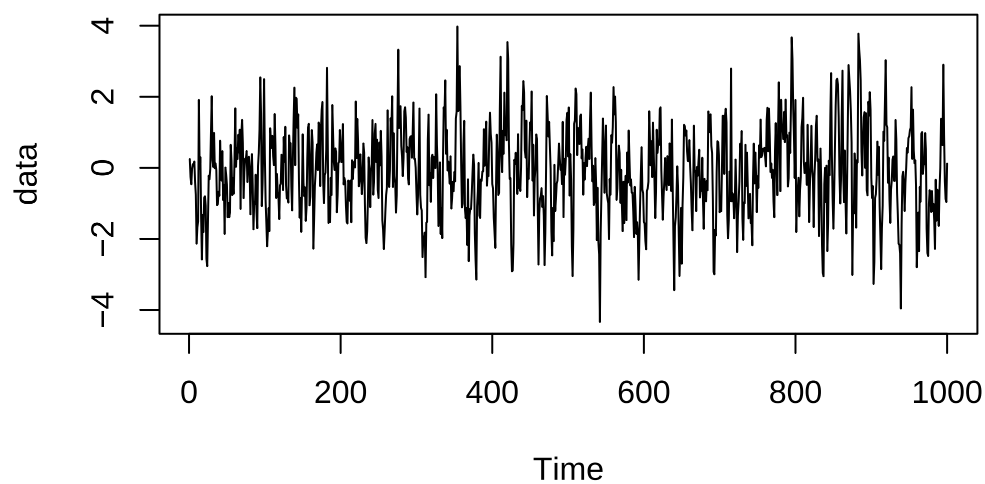
An $MA(q)$ model is invertible if the absolute value of the roots of the characteristic polynomial are larger than 1. Invertible $MA(q)$ processes can be expressed as $AR(\infty)$ processes.

Model Identification

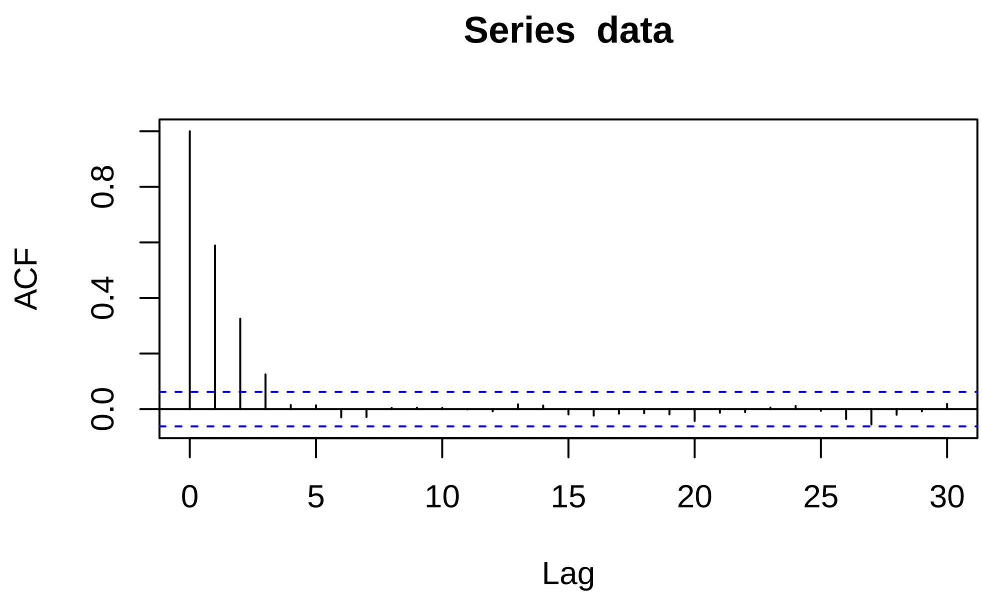
$MA(q)$ models are (weakly) stationary, display clear cut-off in the ACF and PACF displays a fast decaying or a damped sinusoid behaviour.

> If these conditions are met, a MA(q) model may be a good choice.

```
# Let us simulate an MA(3) process
data<-arima.sim(n=1000,list(ma=c(0.6,0.4,0.2)))
plot(data)
```

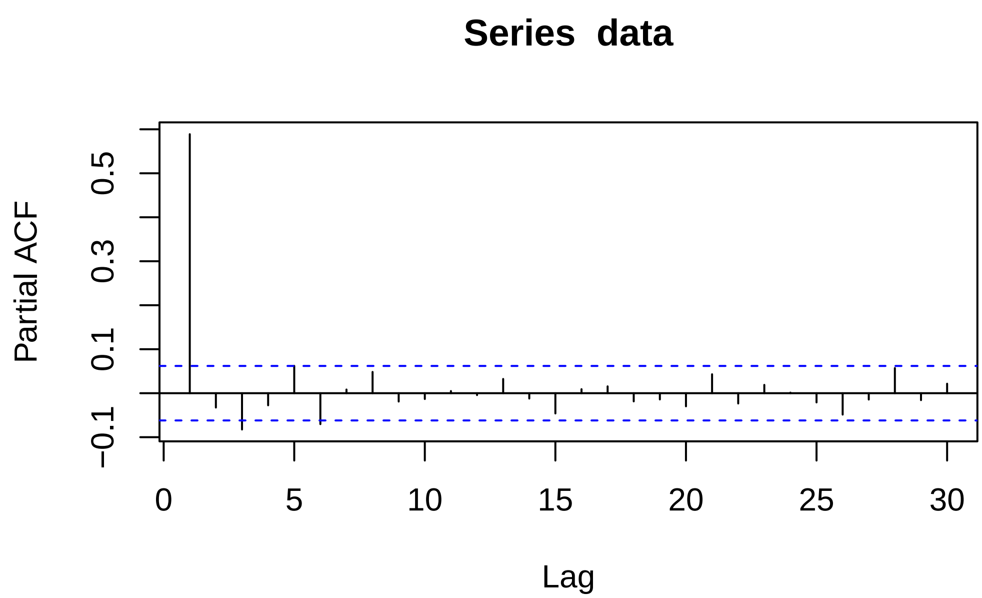



```
acf(data)
```



Series data
ACF
0.0
0.4
0.8
0
5
10
15
20
25
30
Lag


```r
pacf(data)
```



Series data
Partial ACF
−0.1
0.1
0.3
0.5
0
5
10
15
20
25
30
Lag

Parameter Identification

The parameters of an $MA(q)$ model can be estimated with:

- Yule-Walker equations: As in the case of the $AR(p)$ processes, the parameters of $MA(q)$ processes can also be estimated by the Yule-Walker equations (after a global mean has been subtracted).
- Conditional sum of squares: `arima(data, order=c(…), method="CSS")`
- Maximum-likelihood estimation (default method in R): `arima(data, order=c(…),method="CSS-MLE")`

ARMA Models

An $ARMA(p,q)$ process combines an $AR(p)$ and a $MA(q)$ process

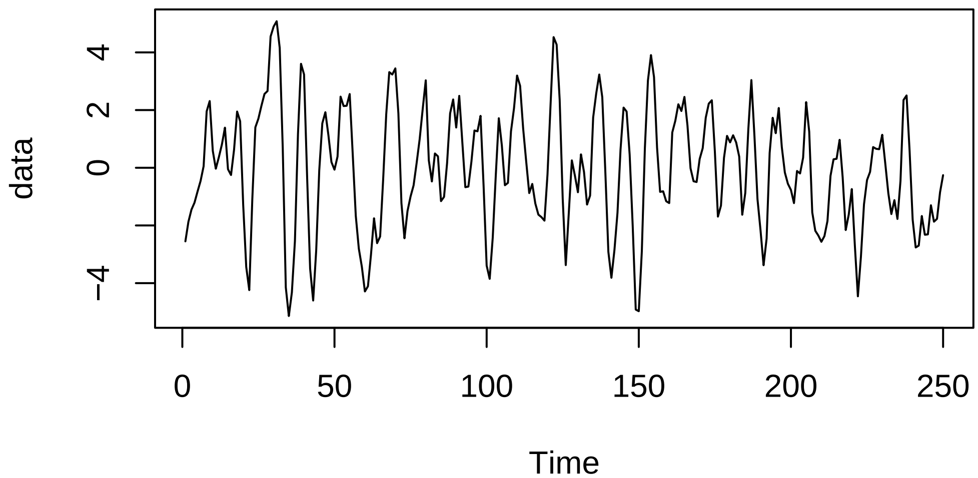
$$X_t = \alpha_1 X_{t-1} + \dots + \alpha_p X_{t-p} + E_t + \beta_1 E_{t-1} + \dots + \beta_q E_{t-q}$$

Definition

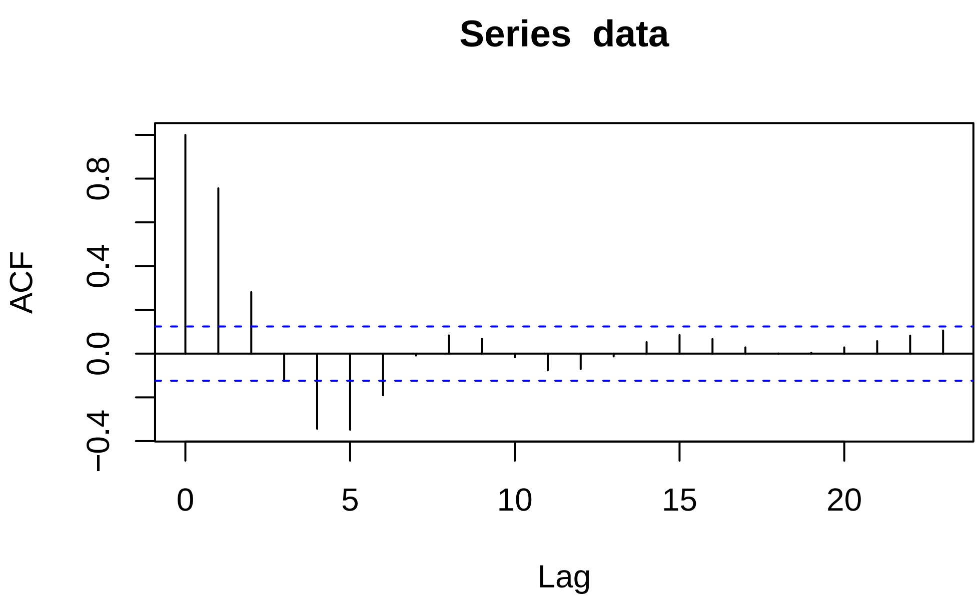
- Stationarity An $ARMA(p,q)$ model is stationary if the absolute value of the roots of characteristic polynom is larger than 1. > Note: Analogy to the $AR(p)$ process
- Invertability An $ARMA(p,q)$ model is invertible if the absolute value of the roots of characteristic polynom is larger than 1. > Note: Analogy to the $MA(q)$ process
- Consequence: Any stationary and invertible $ARMA(p,q)$ model can either be rewritten in the form of an $AR(\infty)$ or an $MA(\infty)$ model.

> **Note**
>
> In practise $ARMA(p,q)$ models are preferred because they require fewer parameters.

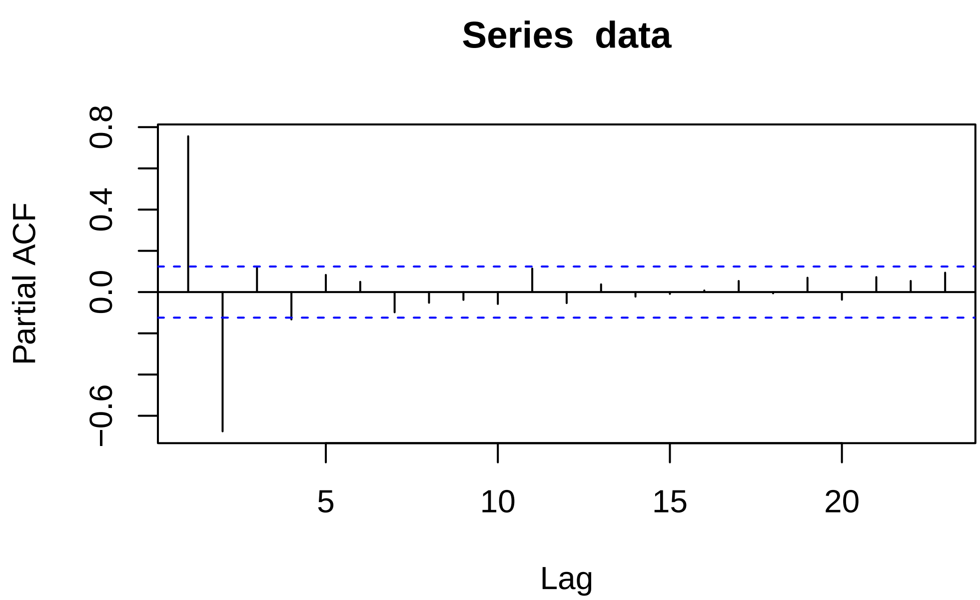
```
data<-arima.sim(n=250,list(ar=c(1.1,-0.5),ma=c(0.4)))
plot(data)
```


data
−4
0
2
4
0
50
100
150
200
250
Time

```r
acf(data)
```


Series data
ACF
−0.4
0.0
0.4
0.8
0
5
10
15
20
Lag

```r
pacf(data)
```



Order Identification

Generic workflow for fitting:

- Is the model stationary?
- Do the properties of (P)ACF match?
- Which orders p,q could be cut off in the (P)ACF?

  Note: Even with theoretical (P)ACF the identification of the cut-off can be challenging.

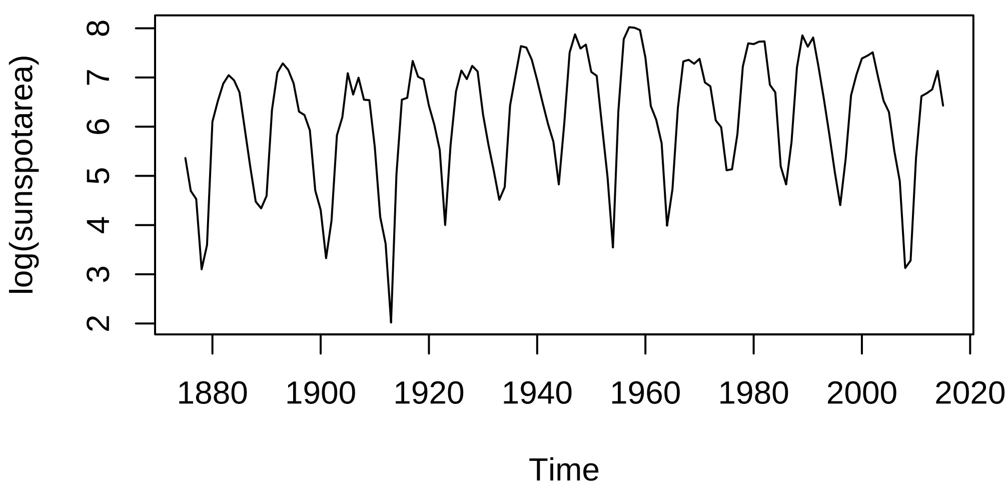
Fitting ARMA Models

- Conditional sum of squares: Applicable for any invertible $ARMA(p, q)$ process `arima(data, order=c(…), method=«CSS»)`
- Maximum likelihood estimation: Performs maximum likelihood estimation after CSS-process `arima(…, order=c(…), method="CSS-ML")`

```r
library(fpp2)
```

```
-- Attaching packages ---------------------------------------------- fpp2 2.5.1 --

v ggplot2   4.0.1     v fma       2.5
v forecast  9.0.2     v expsmooth 2.3
```

```r
plot(log(sunspotarea))
```

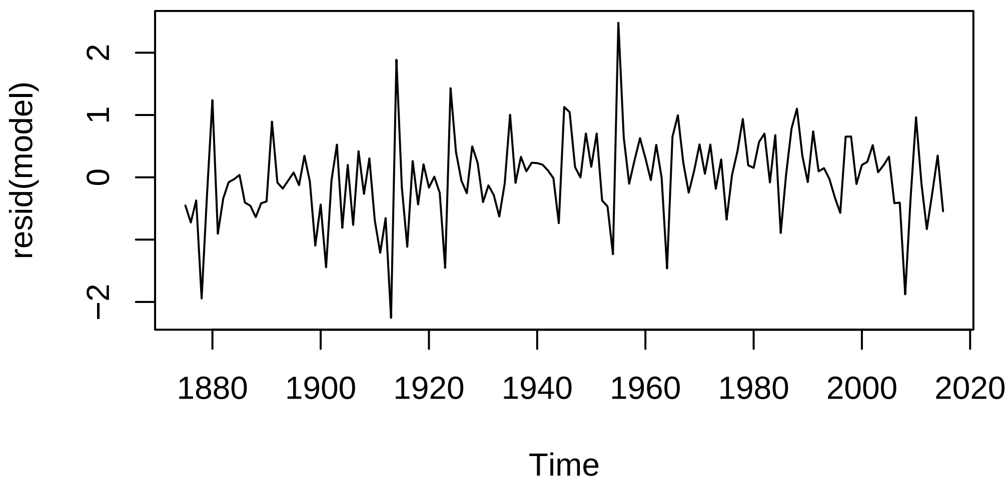



```r
library(forecast)
model <- auto.arima(log(sunspotarea), max.p = 10, max.q = 10, stationary = TRUE)
model
```

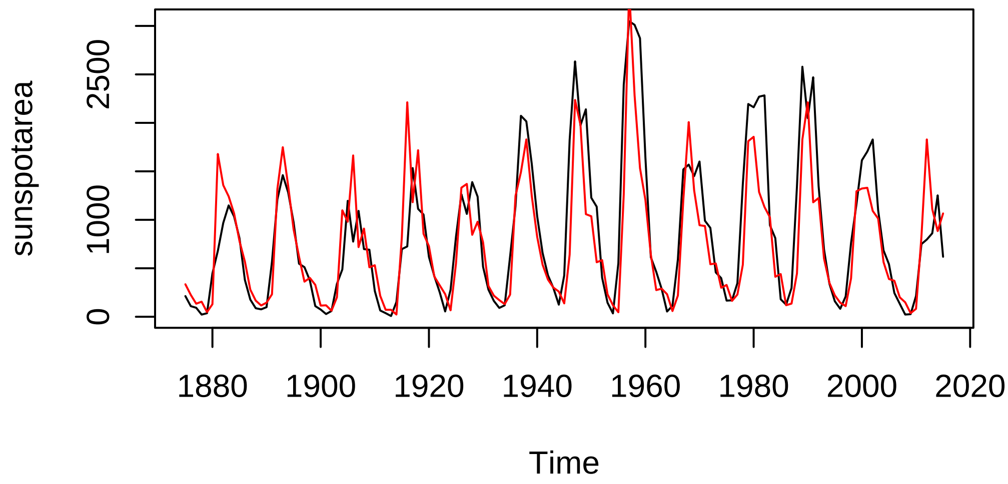
```
Series: log(sunspotarea)
ARIMA(2,0,1) with non-zero mean

Coefficients:
         ar1      ar2      ma1     mean
```

```
      1.4834  -0.7710  -0.5132  6.1885
s.e.  0.0728   0.0605   0.0953  0.0983

sigma^2 = 0.4794:  log likelihood = -147.06
AIC=304.12   AICc=304.56   BIC=318.86
```

```
plot(resid(model))
```




```
plot(sunspotarea)
lines(exp(log(sunspotarea)-resid(model)),col="red")
```



Advantages

Estimates all parameters above and provides error estimates.

Disadvantage

Works for any distribution of the innovation. For distributions that deviate strongly from Gaussian type, results may be unreliable.

Key Properties of Models

Property	AR(p)	MA(q)	ARMA(p,q)
ACF	Fast decay / damped sinusoid	Cut-off at q	Drop at q
PACF	Cut-off at p	Fast decay	Drop at p

Figure 10: Model Properties

W-07-Timeseries-ARMA-GLS

Linear Regression with timeseries

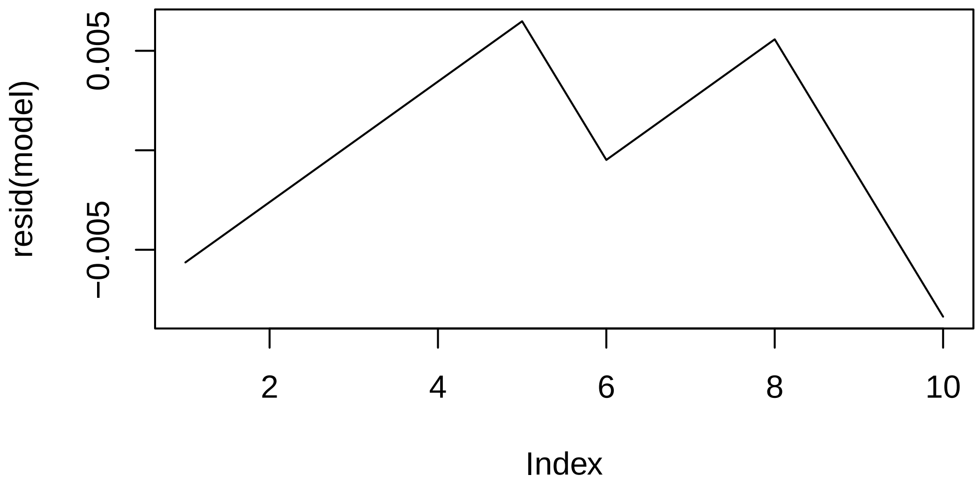
Principle

Model the target time series Y_t as a (linear) combination/function of the constituent, with an error term E_t and the realizations $x_{t1}, \ldots, x_{tp}$ of the p time series. Major challenge: In real-world data, the errors can be strongly correlated.

Find Model Coefficients

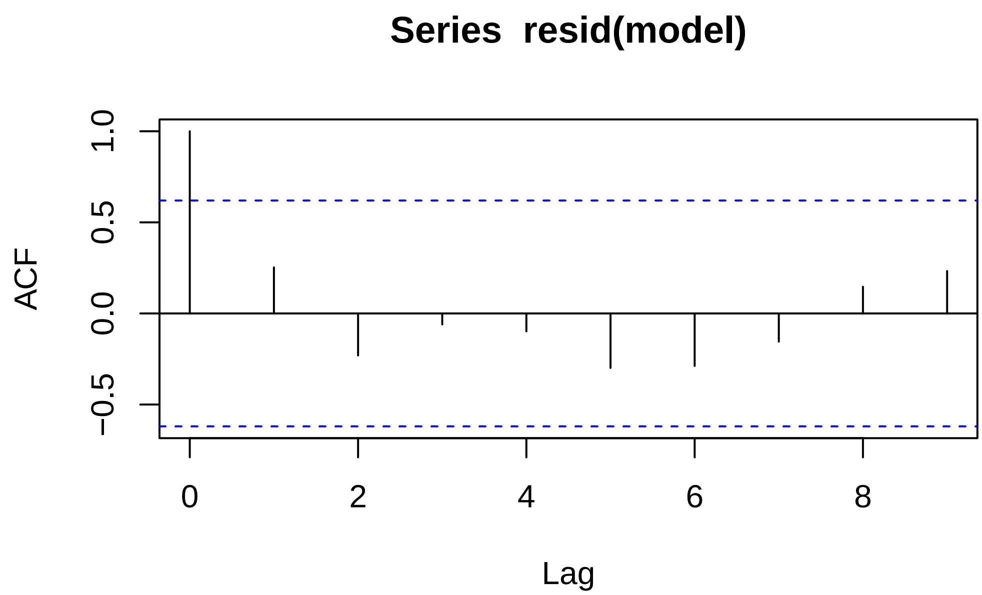
An assumption of linear regression is that the error E_t should be stationary and independent of the input time series. Often it is non-white noise and displays serial correlation. Perform a linear fit on the data and analyze the residuals:

```r
# Sample Data
la_data <- data.frame(
  year = 2010:2019,
  value = c(3.79, 3.82, 3.85, 3.88, 3.91, 3.93, 3.96, 3.99, 4.01, 4.03)
)

# Perform Linear Fit
model <- lm(value ~ year, data = la_data)

plot(resid(model), type="l")
```

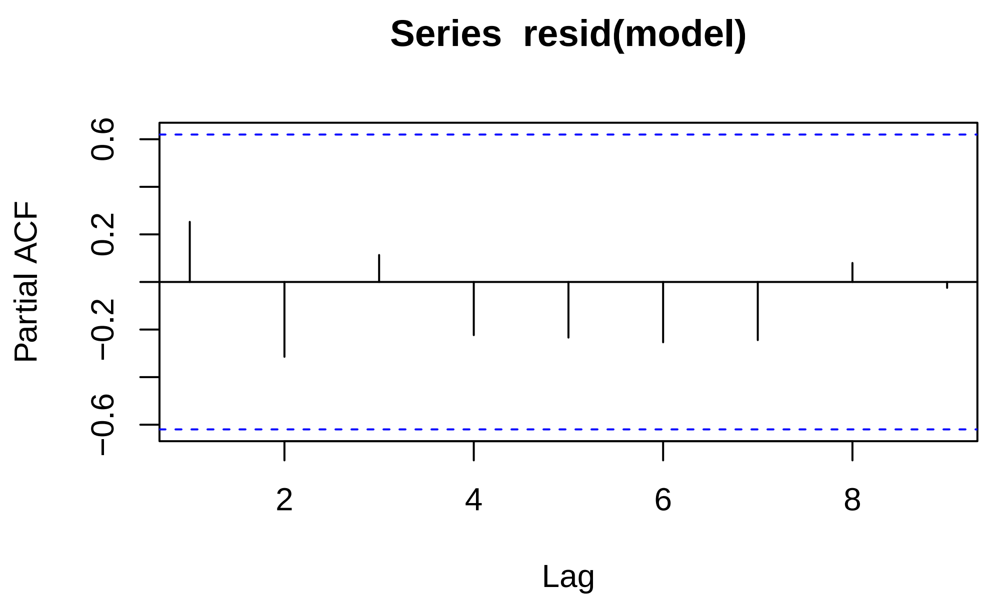


Is there a structure in the residuals?

```
acf(resid(model))
```




```r
pacf(resid(model))
```



Error displays a serial connection and may be modeled by an AR(1) process. Whenever the error terms are dependent, the error estimation of the linear regression is invalid.

> **Note**
>
> Standard deviation is invalid if residuals are correlated

```r
summary(model)
```

```

Call:
lm(formula = value ~ year, data = la_data)

Residuals:
       Min         1Q     Median         3Q        Max 
-0.0083636 -0.0023030 -0.0000303  0.0032273  0.0064848 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -5.041e+01  1.116e+00  -45.19 6.35e-11 ***
year         2.697e-02  5.538e-04   48.70 3.50e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.00503 on 8 degrees of freedom
Multiple R-squared:  0.9966,	Adjusted R-squared:  0.9962 
F-statistic:  2372 on 1 and 8 DF,  p-value: 3.498e-11
```

Durbin-Watson Test

The Durbin-Watson (DW) test is a statistical test used to detect the presence of autocorrelation (specifically first-order serial correlation) in the residuals from a regression analysis.

In a standard linear regression, we assume that residuals are independent. If they are correlated (e.g., if the error at time t is related to the error at time $t-1$), your standard errors may be underestimated, leading to misleadingly "significant" results.

```r
library(lmtest)
```

```
Loading required package: zoo
```

```

Attaching package: 'zoo'
```

```
The following objects are masked from 'package:base':

    as.Date, as.Date.numeric
```

```r
dwtest(model)
```

```

	Durbin-Watson test

data:  model
DW = 0.99211, p-value = 0.009606
alternative hypothesis: true autocorrelation is greater than 0
```

Cochrane-Orcutt

The Cochrane-Orcutt method is an iterative procedure used in econometrics and statistics to estimate a linear regression model in the presence of autocorrelation (specifically AR(1) errors).

When the Durbin-Watson test indicates that residuals are correlated, the standard Ordinary Least Squares (OLS) assumptions are violated. Cochrane-Orcutt "cleanses" the model by transforming the variables to eliminate this correlation.

Steps

1. Perform ordinary least square fit on data.
2. Analyse (P)ACF and fit AR(1) model on residuals.
3. Transform data according to parameter of the residuals' AR(1) process.
4. Fit the transformed data to determine the parameters of the linear model.

Generalized Least Square Methode

Generalized Least Squares (GLS) is a powerful extension of Ordinary Least Squares (OLS) used to estimate unknown parameters in a linear regression model when the underlying assumptions about the error terms are violated. Specifically, it is used when there is heteroscedasticity (non-constant variance) or autocorrelation (correlated errors).

Impact of Missing Variables

Correlated errors are often caused by missing input variables. In a perfect world, these parameters would be included in the model. However, often it is impossible to provide the input.

W-08-Timeseries-GLS-ARIMA

An ARIMA model is basically an ARMA model that has been “stretched” to include a trend. Because it contains that trend (the unit root), it doesn’t settle back to a constant average, making it non-stationary.

Definition

ARIMA stands for AutoRegressive Integrated Moving Average. It is a powerful and flexible class of statistical models used for analyzing and forecasting time-series data. Unlike simple linear regression, ARIMA explicitly accounts for the dependencies between observations over time. The model is defined by three parameters p, d, q.

Interpreting the Model Output

```r
library(TSA)
```

```
Registered S3 methods overwritten by 'TSA':
  method       from
  fitted.Arima forecast
  plot.Arima   forecast


Attaching package: 'TSA'


The following objects are masked from 'package:stats':

    acf, arima


The following object is masked from 'package:utils':

    tar
```

```r
data(oil.price)
fit<-arima(log(oil.price),order=c(2,1,1))
fit
```

```
Call:
arima(x = log(oil.price), order = c(2, 1, 1))

Coefficients:
          ar1     ar2     ma1
      -0.3240  0.0150  0.5981
s.e.   0.2653  0.1107  0.2571

sigma^2 estimated as 0.006642:  log likelihood = 261.12,  aic = -516.25
```

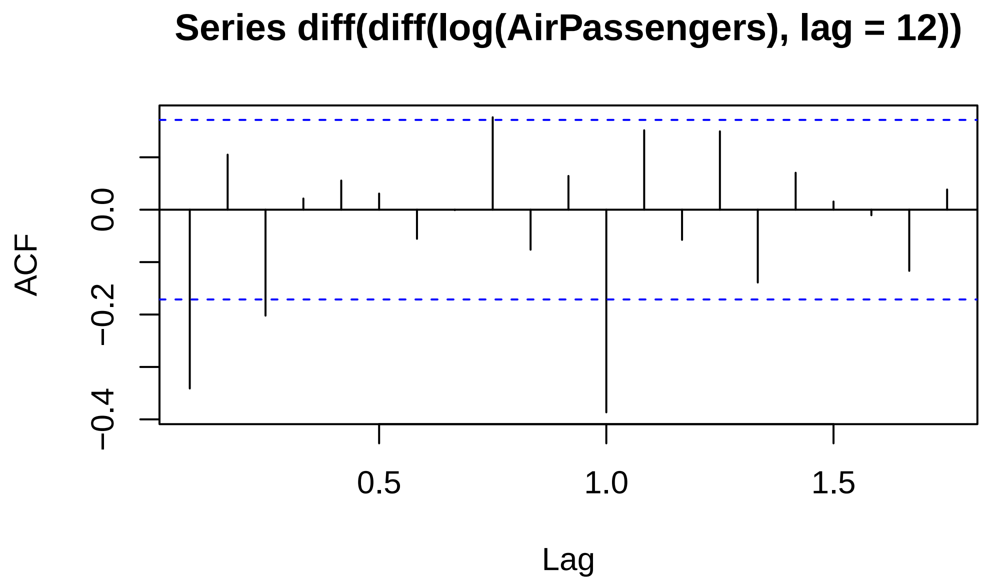
This ARIMA(2,1,1) model is equal to an ARMA(3,1) model.

W-09-Timeseries-ARIMA-SARIMA

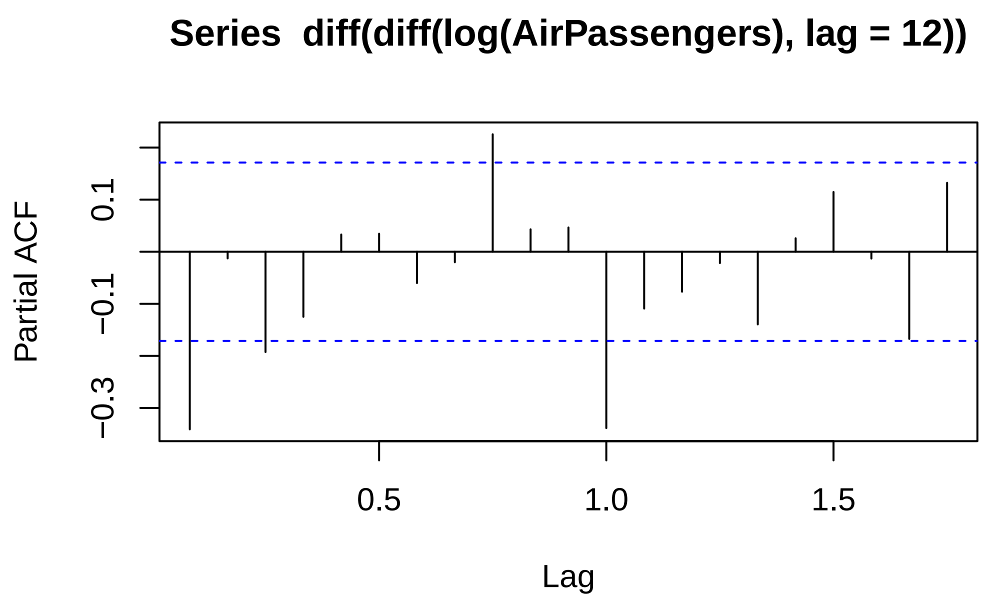
Sarima Models

SARIMA stands for **S**easonal **A**utoregressive **I**ntegrated **M**oving **A**verage. It is an extension of the standard ARIMA model specifically designed to handle univariate time series data that contains a seasonal component. While ARIMA focuses on the relationship between points in time (trend), SARIMA adds a layer to account for cycles that repeat over specific intervals (e.g., monthly sales spikes or daily temperature changes).

```r
data("AirPassengers")
acf(diff(diff(log(AirPassengers),lag=12)))
```



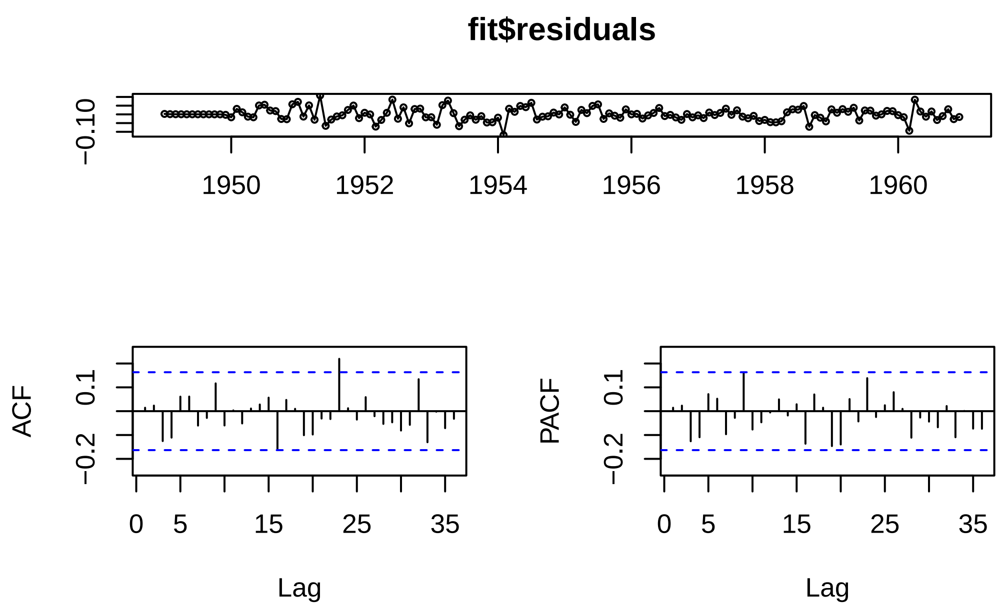

```r
pacf(diff(diff(log(AirPassengers),lag=12)))
```



```r
fit<-arima(log(AirPassengers),order=c(0,1,1),seasonal=c(0,1,1))
fit
```

```
Call:
arima(x = log(AirPassengers), order = c(0, 1, 1), seasonal = c(0, 1, 1))

Coefficients:
          ma1     sma1
      -0.4018  -0.5569
s.e.   0.0896   0.0731

sigma^2 estimated as 0.001348:  log likelihood = 244.7,  aic = -485.4
```

```r
library(forecast)
tsdisplay(fit$residuals)
```




```r
fitAuto<-auto.arima(log(AirPassengers),ic="aic")
fitAuto
```

```
Series: log(AirPassengers)
```

```
ARIMA(0,1,1)(0,1,1)[12]

Coefficients:
          ma1     sma1
      -0.4018  -0.5569
s.e.   0.0896   0.0731

sigma^2 = 0.001371:  log likelihood = 244.7
AIC=-483.4   AICc=-483.21   BIC=-474.77
```

> Note: The automated procedure selects the same model.

How to fit SARIMA Models?

1. Estimate the seasonality period s visually. The order of differentiation D (for the seasonality part) is typically less or equal to 1.
2. Check if another differentiation of lag 1 is required for stationarity. If yes, try d=1, else d=0.
3. Analyse (P)ACF of the result of the two differentiations. Determine p,q from the short-lag behavior and P,Q from the multiple-of-the-period dependency.
4. Fit the model with arima(data,order=c(p,d,q),seasonal=c(P,D,Q))
5. Check the applicability of the model by analyzing the residuals (residuals should behave like (Gaussian) white noise).

Non-Linear Models

Linear models are models that can be formulated as linear combination of different time series elements X_t and errors/innovations E_t.

> Note: An example are data with changing standard deviation.

Heteroscedasticity

Heteroscedasticity occurs when the variance of the residuals (errors) in a model is not constant across all levels of the independent variables or across time.

> Note: Heteroskedastic time series are not stationary, because the variance is not equal for all time points.

ARCH Models

A time series is autoregressive conditional heteroskedastic of the order p (abbreviated ARCH(p)) if it has the form:

$$X_t = \sigma_t W_t \quad \text{with} \quad \sigma_t = \sqrt{\alpha_0 + \sum_{i=1}^{p} \alpha_i X_{t-i}^2}$$

Fitting ARCH Models

```r
#load SMI data
data("EuStockMarkets")

esm <- EuStockMarkets
tmp <- EuStockMarkets[,2]
smi <- ts(tmp, start=start(esm), freq=frequency(esm))
lret.smi <- diff(log(smi))

#perform fit
library(tseries)
```

```
Registered S3 method overwritten by 'quantmod':
  method            from
  as.zoo.data.frame zoo
```

```r
fit<-garch(lret.smi, order=c(0,2))
```

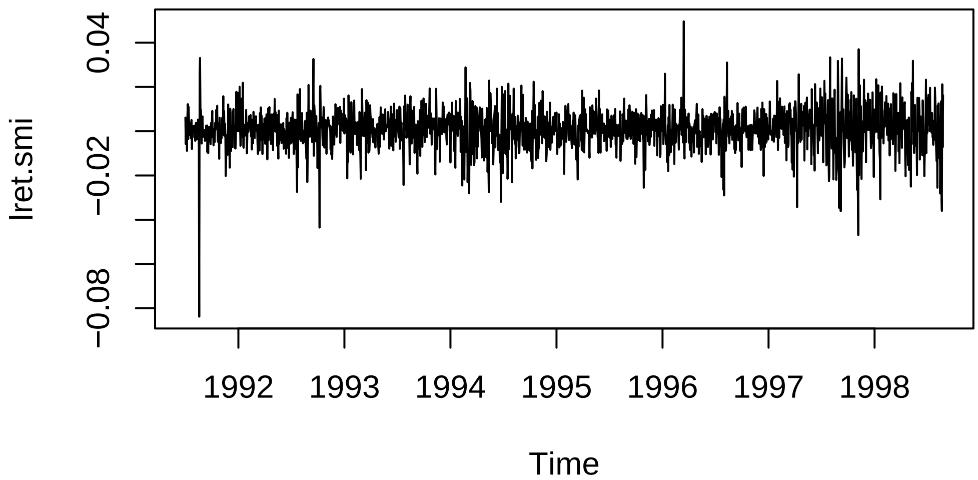
```

 ***** ESTIMATION WITH ANALYTICAL GRADIENT *****

     I     INITIAL X(I)        D(I)

     1     7.700685e-05     1.000e+00
     2     5.000000e-02     1.000e+00
     3     5.000000e-02     1.000e+00

    IT   NF      F         RELDF    PRELDF    RELDX   STPPAR   D*STEP   NPRELDF
     0    1 -7.796e+03
     1    8 -7.797e+03  7.76e-05  1.39e-04  2.7e-05  1.4e+11  2.7e-06  9.94e+06
```

```
   2     9 -7.797e+03  7.84e-07  8.41e-07  2.1e-05  2.0e+00  2.7e-06  3.10e+00
   3    17 -7.803e+03  8.48e-04  1.27e-03  2.6e-01  2.0e+00  4.5e-02  3.10e+00
   4    19 -7.803e+03  5.19e-06  7.74e-06  2.0e-02  2.0e+00  4.5e-03  1.44e-03
   5    20 -7.803e+03  2.15e-07  2.70e-06  2.0e-02  2.0e+00  4.5e-03  3.72e-04
   6    21 -7.803e+03  3.22e-06  2.03e-05  1.3e-02  1.9e+00  2.2e-03  3.91e-04
   7    24 -7.805e+03  1.73e-04  1.62e-04  9.7e-02  4.9e-01  2.1e-02  1.87e-04
   8    25 -7.805e+03  9.03e-05  1.05e-04  1.6e-01  0.0e+00  3.8e-02  1.05e-04
   9    26 -7.805e+03  1.19e-05  1.35e-05  4.7e-02  0.0e+00  1.3e-02  1.35e-05
  10    27 -7.805e+03  1.16e-07  1.07e-07  4.1e-03  0.0e+00  1.1e-03  1.07e-07
  11    28 -7.805e+03  1.13e-09  1.10e-09  3.2e-04  0.0e+00  1.0e-04  1.10e-09
  12    29 -7.805e+03  8.85e-12  7.83e-12  4.0e-05  0.0e+00  1.1e-05  7.83e-12

***** RELATIVE FUNCTION CONVERGENCE *****

FUNCTION    -7.805450e+03   RELDX        3.982e-05
FUNC. EVALS      29         GRAD. EVALS      13
PRELDF       7.826e-12      NPRELDF      7.826e-12

    I      FINAL X(I)        D(I)          G(I)

    1    6.567582e-05     1.000e+00     1.512e+00
    2    1.309070e-01     1.000e+00     1.394e-03
    3    1.074336e-01     1.000e+00    -8.848e-04
```

```
fit$coef
```

```
          a0           a1           a2 
6.567582e-05 1.309070e-01 1.074336e-01 
```

```
plot(lret.smi)
```




```r
fit$fitted.values
```

```
Time Series:
Start = c(1991, 131)
End = c(1998, 169)
Frequency = 260
                sigt        -sigt
1991.500          NA           NA
1991.504          NA           NA
1991.508 0.008619948 -0.008619948
1991.512 0.008413776 -0.008413776
1991.515 0.008192281 -0.008192281
1991.519 0.008738385 -0.008738385
1991.523 0.008954976 -0.008954976
1991.527 0.009492994 -0.009492994
1991.531 0.009133273 -0.009133273
1991.535 0.009107082 -0.009107082
1991.538 0.009010984 -0.009010984
1991.542 0.008347955 -0.008347955
1991.546 0.008228813 -0.008228813
1991.550 0.008218464 -0.008218464
1991.554 0.008182066 -0.008182066
```

1991.558	0.008192224	-0.008192224
1991.562	0.008379568	-0.008379568
1991.565	0.008776044	-0.008776044
1991.569	0.008527705	-0.008527705
1991.573	0.008162093	-0.008162093
1991.577	0.008164387	-0.008164387
1991.581	0.008160428	-0.008160428
1991.585	0.008151230	-0.008151230
1991.588	0.008112088	-0.008112088
1991.592	0.008156176	-0.008156176
1991.596	0.008296850	-0.008296850
1991.600	0.008366156	-0.008366156
1991.604	0.008332257	-0.008332257
1991.608	0.008198592	-0.008198592
1991.612	0.008183117	-0.008183117
1991.615	0.008274863	-0.008274863
1991.619	0.008207454	-0.008207454
1991.623	0.008337280	-0.008337280
1991.627	0.008361690	-0.008361690
1991.631	0.008594171	-0.008594171
1991.635	0.031486098	-0.031486098
1991.638	0.030248146	-0.030248146
1991.642	0.016931529	-0.016931529
1991.646	0.013959892	-0.013959892
1991.650	0.009355178	-0.009355178
1991.654	0.008882279	-0.008882279
1991.658	0.008496063	-0.008496063
1991.662	0.008529784	-0.008529784
1991.665	0.008638284	-0.008638284
1991.669	0.008397910	-0.008397910
1991.673	0.008152679	-0.008152679
1991.677	0.008146693	-0.008146693
1991.681	0.008157023	-0.008157023
1991.685	0.008147135	-0.008147135
1991.688	0.008261294	-0.008261294
1991.692	0.008293718	-0.008293718
1991.696	0.008277651	-0.008277651
1991.700	0.008255187	-0.008255187
1991.704	0.008146510	-0.008146510
1991.708	0.008159992	-0.008159992
1991.712	0.008849384	-0.008849384
1991.715	0.008685582	-0.008685582
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1998.115	0.010272206	-0.010272206
1998.119	0.009792821	-0.009792821
1998.123	0.008284676	-0.008284676
1998.127	0.008802266	-0.008802266
1998.131	0.009772739	-0.009772739
1998.135	0.009205073	-0.009205073
1998.138	0.008396372	-0.008396372
1998.142	0.009048822	-0.009048822
1998.146	0.009630320	-0.009630320
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1998.158	0.009244115	-0.009244115
1998.162	0.009133388	-0.009133388
1998.165	0.008211898	-0.008211898
1998.169	0.008311122	-0.008311122

1998.173	0.008255823	-0.008255823
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1998.185	0.010236198	-0.010236198
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1998.196	0.009788478	-0.009788478
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1998.215	0.008638300	-0.008638300
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1998.265	0.010123122	-0.010123122
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1998.315	0.008562413	-0.008562413
1998.319	0.010030737	-0.010030737
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1998.335	0.010384665	-0.010384665

1998.338	0.008806126	-0.008806126
1998.342	0.008498968	-0.008498968
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1998.485	0.009043701	-0.009043701
1998.488	0.011865519	-0.011865519
1998.492	0.011695373	-0.011695373
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```
1998.504 0.009744095 -0.009744095
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1998.638 0.018156125 -0.018156125
1998.642 0.016252704 -0.016252704
1998.646 0.010992686 -0.010992686
```
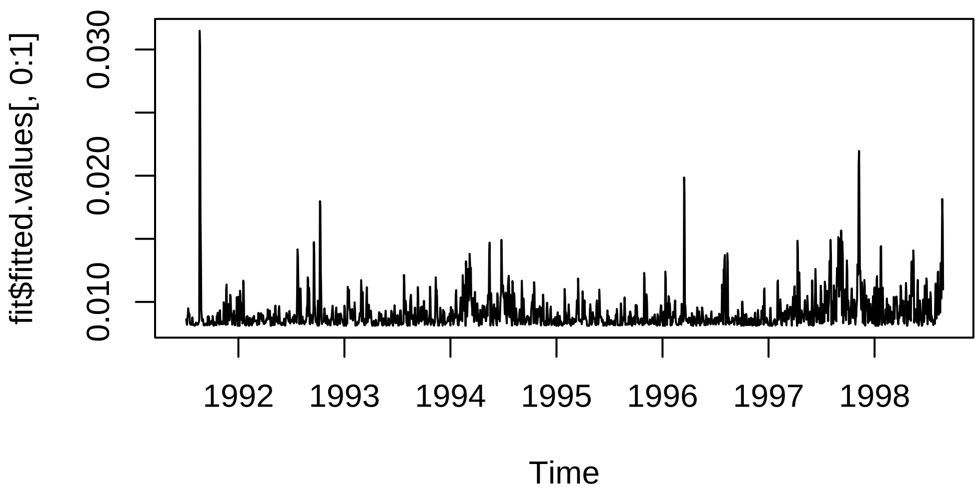
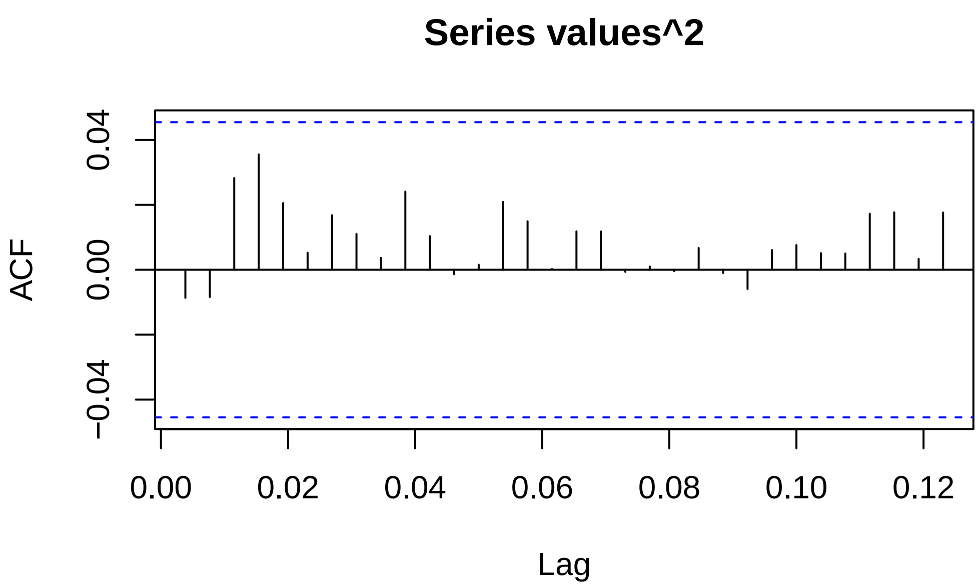
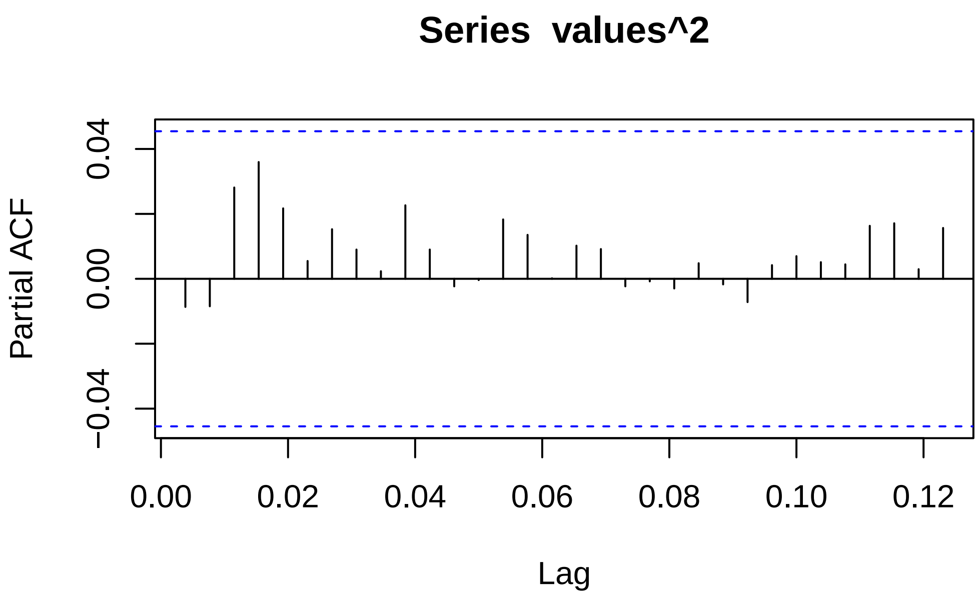

```
plot(fit$fitted.values[,0:1])
```




```
values<-fit$residuals
acf(values^2,na.action=na.pass)
```



```r
pacf(values^2,na.action=na.pass)
```



Fundamentals of Forecasting

Generally speaking, we are looking for the most likely next k values based on the first n observations.

Sources of Uncertainty in Forecasting

Principal sources of uncertainty:

1. Is the model of the past data also appropriate for the future (e.g. change of behavior, policy)?
2. Did you choose the order of the ARMA(p,q) correctly?
3. Are the fitting parameters in the model (sufficiently) accurate?
4. How can you assess the contribution/variability of the innovation terms E_t?

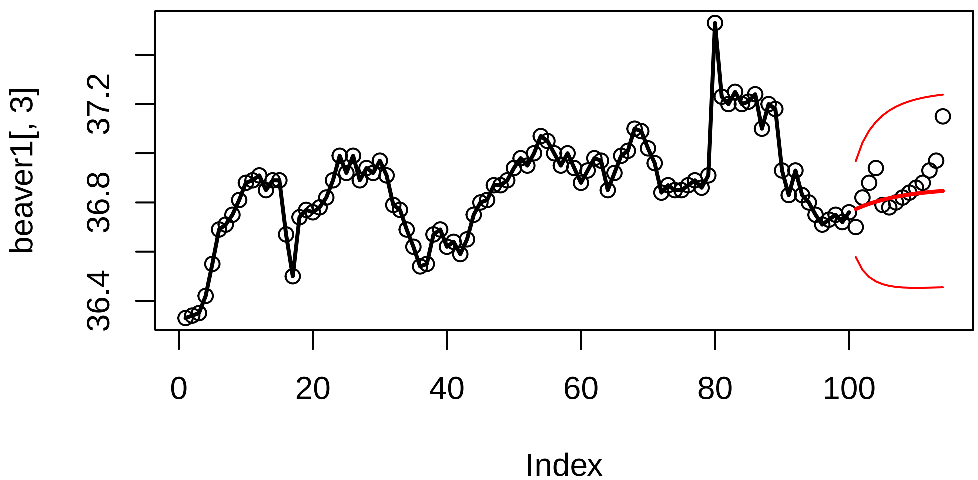
Forecasting different Models

$AR(1)$ processes

For an $AR(1)$ model the forecasted value is just the last value times the fitting coefficient α_1.

> Note: Only the previous value is required.

```
#Extract training set
btrain<-window(beaver1[,3],1,100)
#Fit AR(1) model with Burg's algorithm
fit<-ar.burg(btrain,order=1)
#Perform forecast of 14 steps
forecast <- predict(fit, n.ahead=14)
#Plot forecast
plot(beaver1[,3], lty=3)
lines(btrain, lwd=2)
lines(forecast$pred, lwd=2, col="red")
lines(forecast$pred+forecast$se*1.96, col="red")
lines(forecast$pred-forecast$se*1.96, col="red")
```



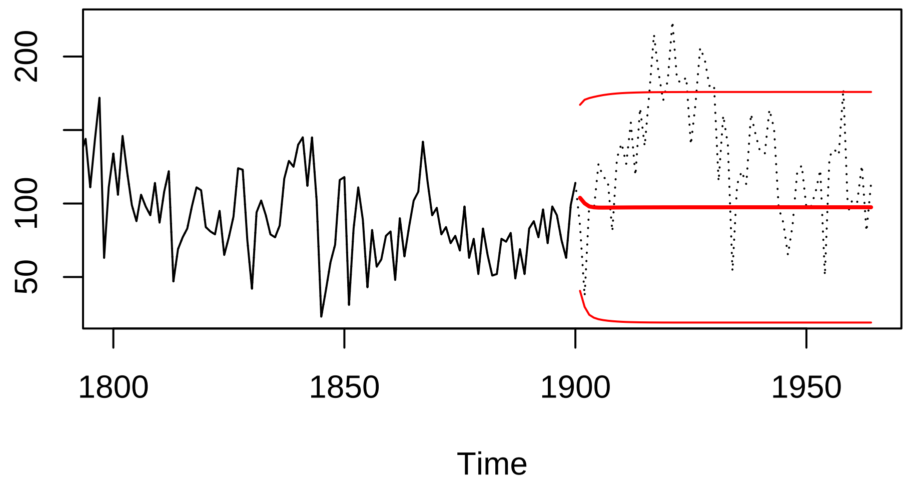
$AR(p)$ **models**

For an $AR(p)$ model the p-step forecast is k steps ahead, the formula is calculated with the previous predictions. The prognosis intervals only consider the uncertainty of the (Gaussian) innovation. Other sources are neglected.

Forecastin $ARIMA$ Models

Mathematically derivation of prediction more involved. However, everything is implemented in the predict function.

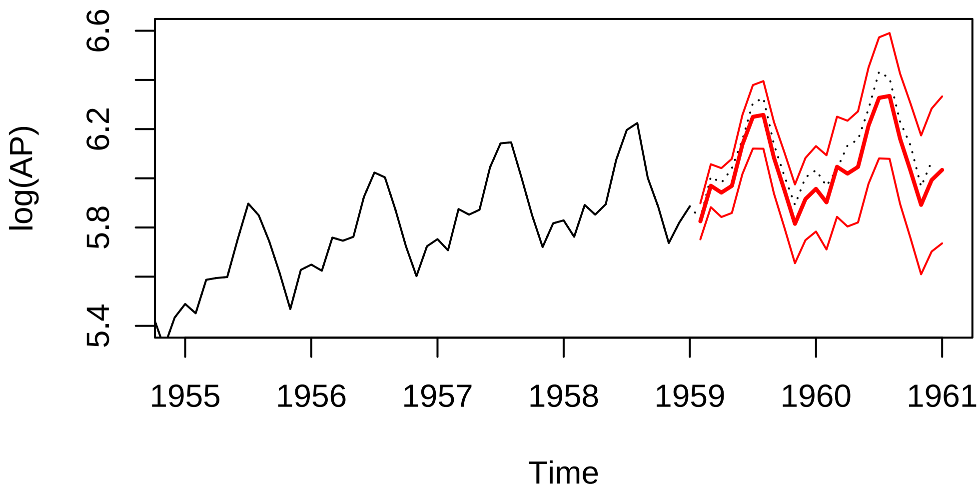
```
#Read data
t.url <- "http://stat.ethz.ch/Teaching/Datasets/WBL/foehre.dat"
douglasfir <- ts(scan(t.url, skip=1), start=1107)
train <- window(douglasfir, start=1107, end=1900)

#Generate ARMA(4,1) model
fit <- arima(train, order=c(4,0,1))

# Predict 64 steps
fc <- predict(fit, n.ahead=64)
# Plot result
plot(window(douglasfir, 1800, 1964), lty=3, ylab="")
lines(train, lwd=1)
lines(fc$pred, lwd=2, col="red")
lines(fc$pred+fc$se*1.96, col="red")
lines(fc$pred-fc$se*1.96, col="red")
```



Forecast for time series with trend and seasonality

If data are fit by an $SARIMA$ model, R takes care of all aspects of the forecast.

```
#Load data
lap<-log(AirPassengers)
ltrain<-window(lap,1949,1959)

#Perform fit
fit<-arima(ltrain,order=c(0,1,1),seasonal=c(0,1,1))

#Perform forecast
fc <- predict(fit, n.ahead=24)

#Plot everything
plot(lap, lty=3,
xlim=c(1955,1961),ylim=c(5.4,6.6),ylab="log(AP)")
lines(ltrain, lwd=1)
lines(fc$pred, lwd=2, col="red")
lines(fc$pred+fc$se*1.96, col="red")
lines(fc$pred-fc$se*1.96, col="red")
```



Features

- The predicted values capture the trend and seasonality.
- The prediction intervals will typically increase over time.
- The prediction quality of the trend is not ascertained and may be poor.